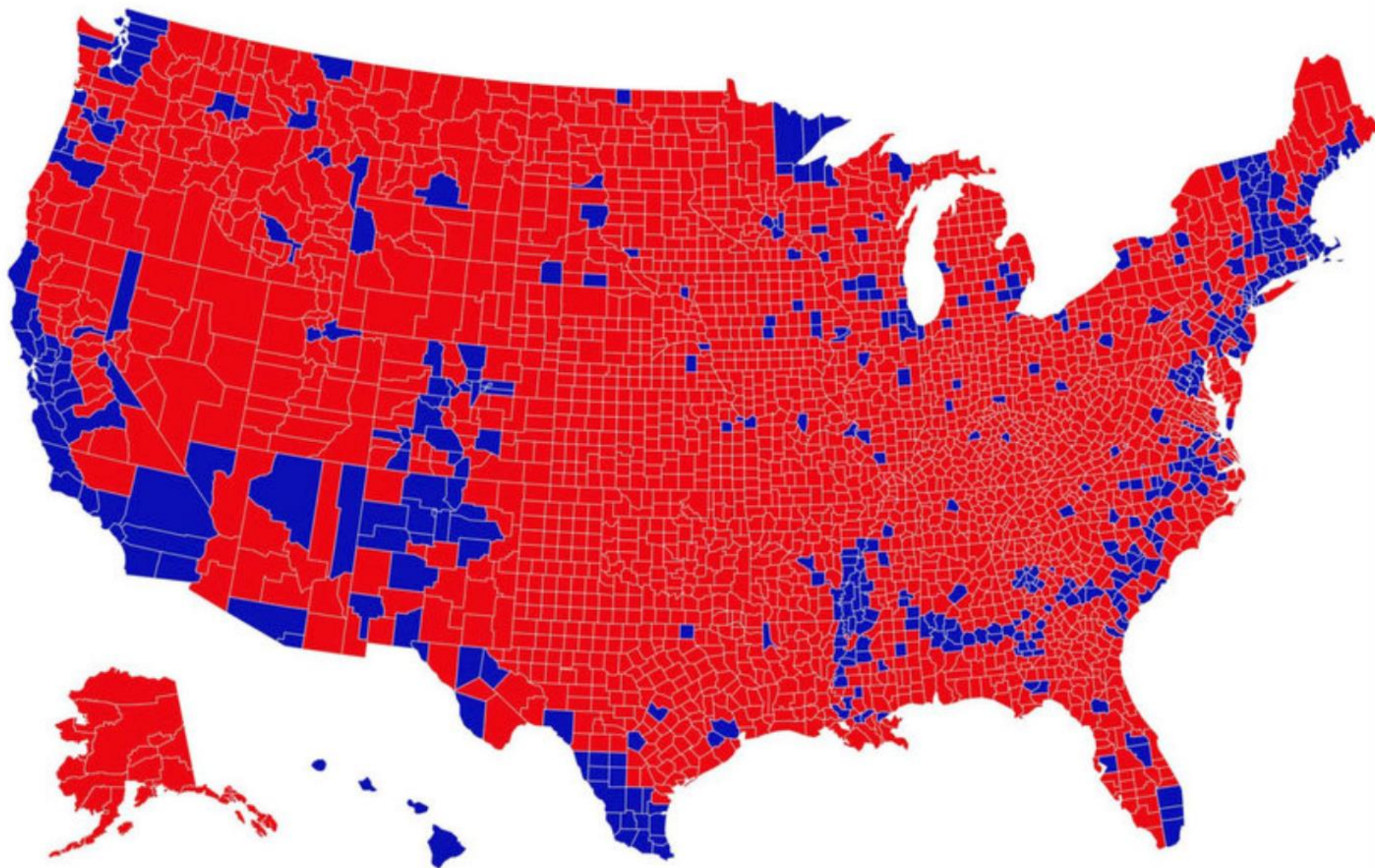


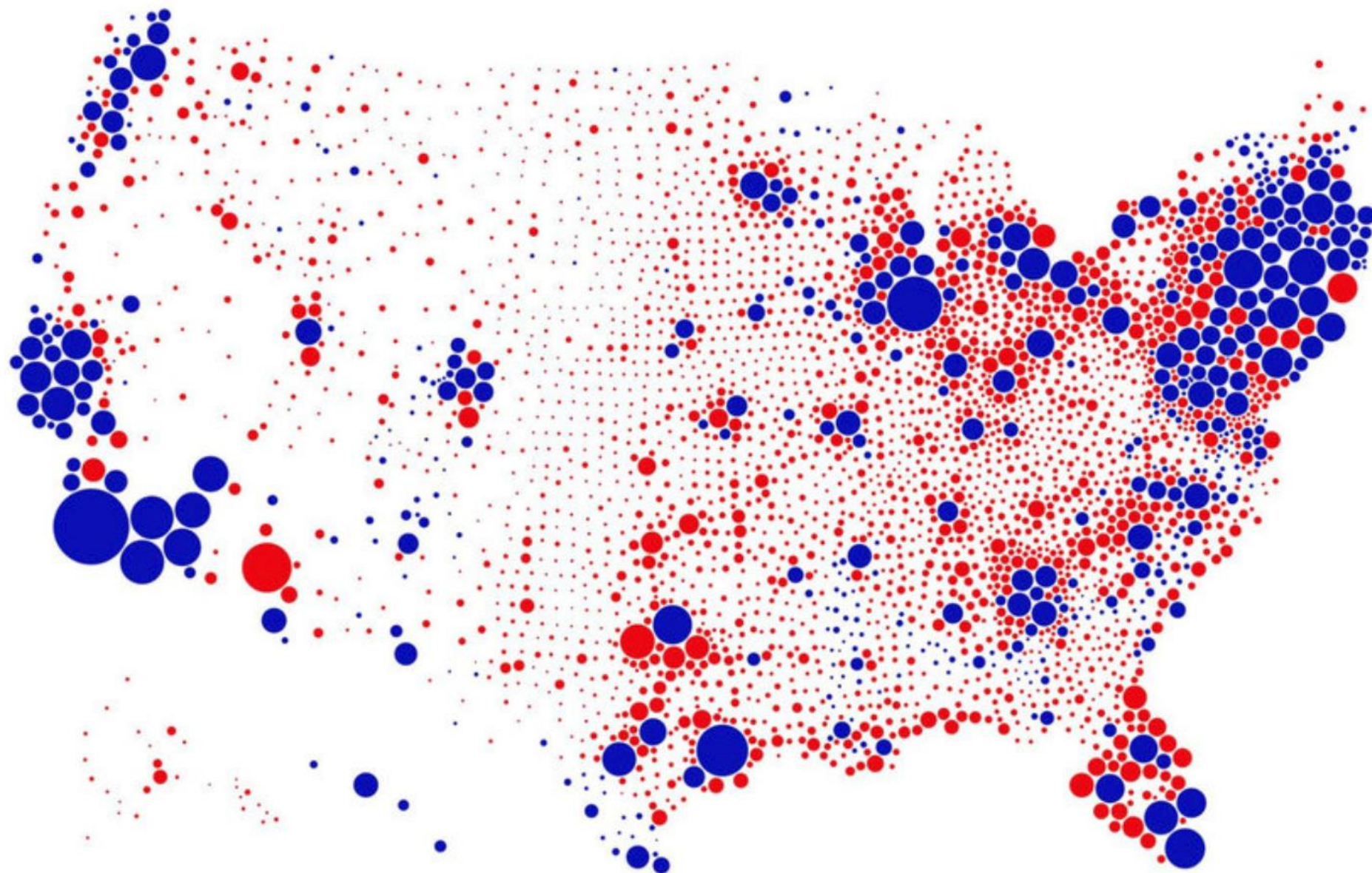
Visualization Ethics

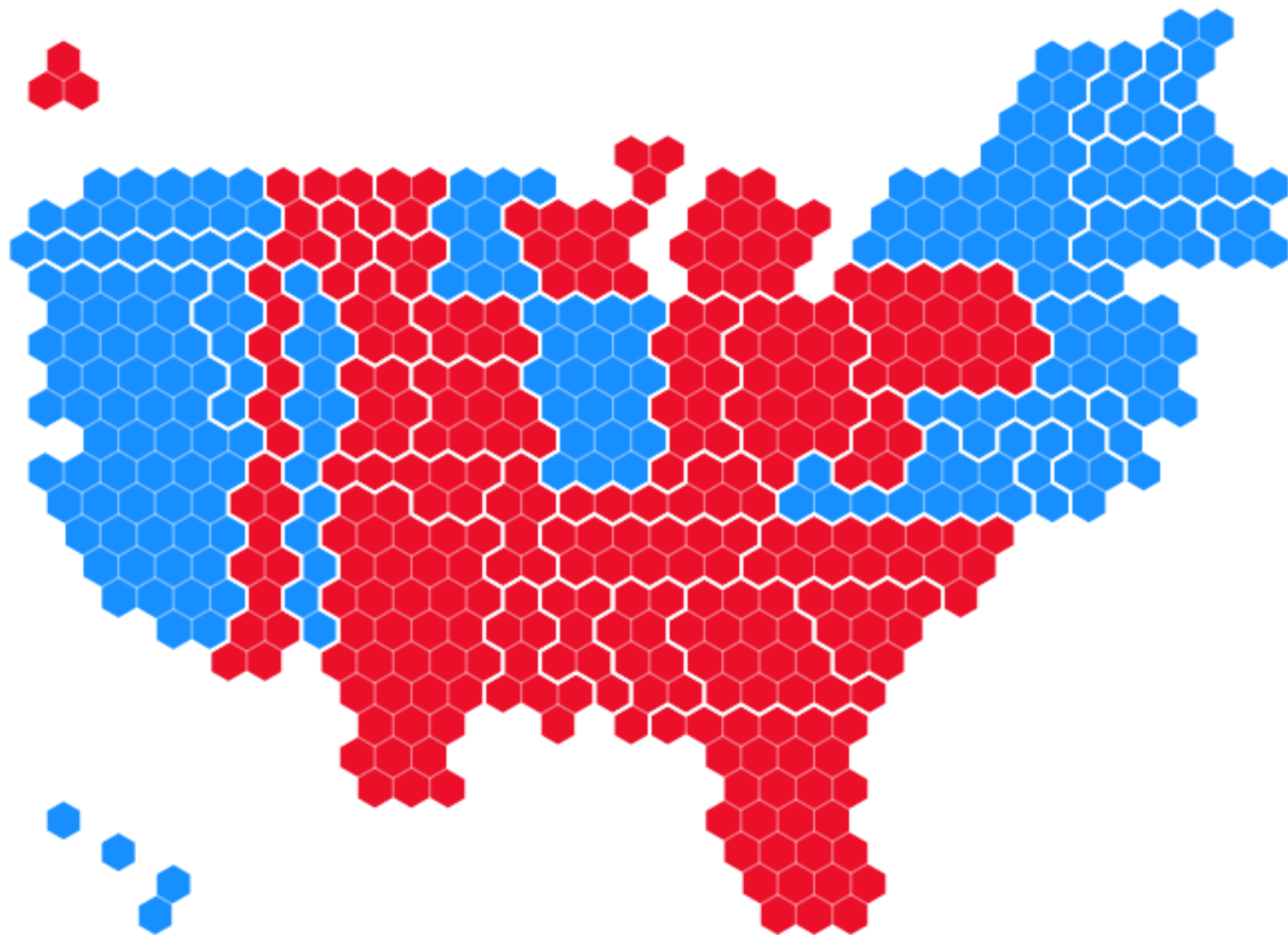
HUGEN 2073

Genomic Data Visualization & Integration

Adapted from Ryan L. Minster, PhD, MSIS with permission







Learning Objectives

By the end of the session, students will be able to:

- Describe common ways that visualizations can mislead
- Employ strategies to prevent the use of visualizations that mislead
- Use basic techniques that enhance the accessibility of visualizations
- Describe how visualizations can be designed to be more equitable and inclusive

Ethics

- **Veracity** truth, accuracy
- **Fidelity** integrity in relationships, keeping promises
- **Autonomy** self-agency, ability to make decisions for oneself
- **Justice** fairness, equity, non-discrimination, transparency
- **Nonmaleficence** avoid harm
- **Beneficence** do good

Good charts that are correctly read enable conversations. Bad charts—or charts that misinterpreted—hinder those conversations.

Alberto Cairo
How Charts Lie

Visualizations as a Conversation

- Explanation
- Persuasion

Common Ways that Visualizations Mislead

- **Fabrication**

Common Ways that Visualizations Mislead

- **Fabrication**
- **Omission**
 - Deletion of inconvenient points
 - Deletion of extreme values
 - Smoothing
 - Aggregation

JOB LOSS BY QUARTER



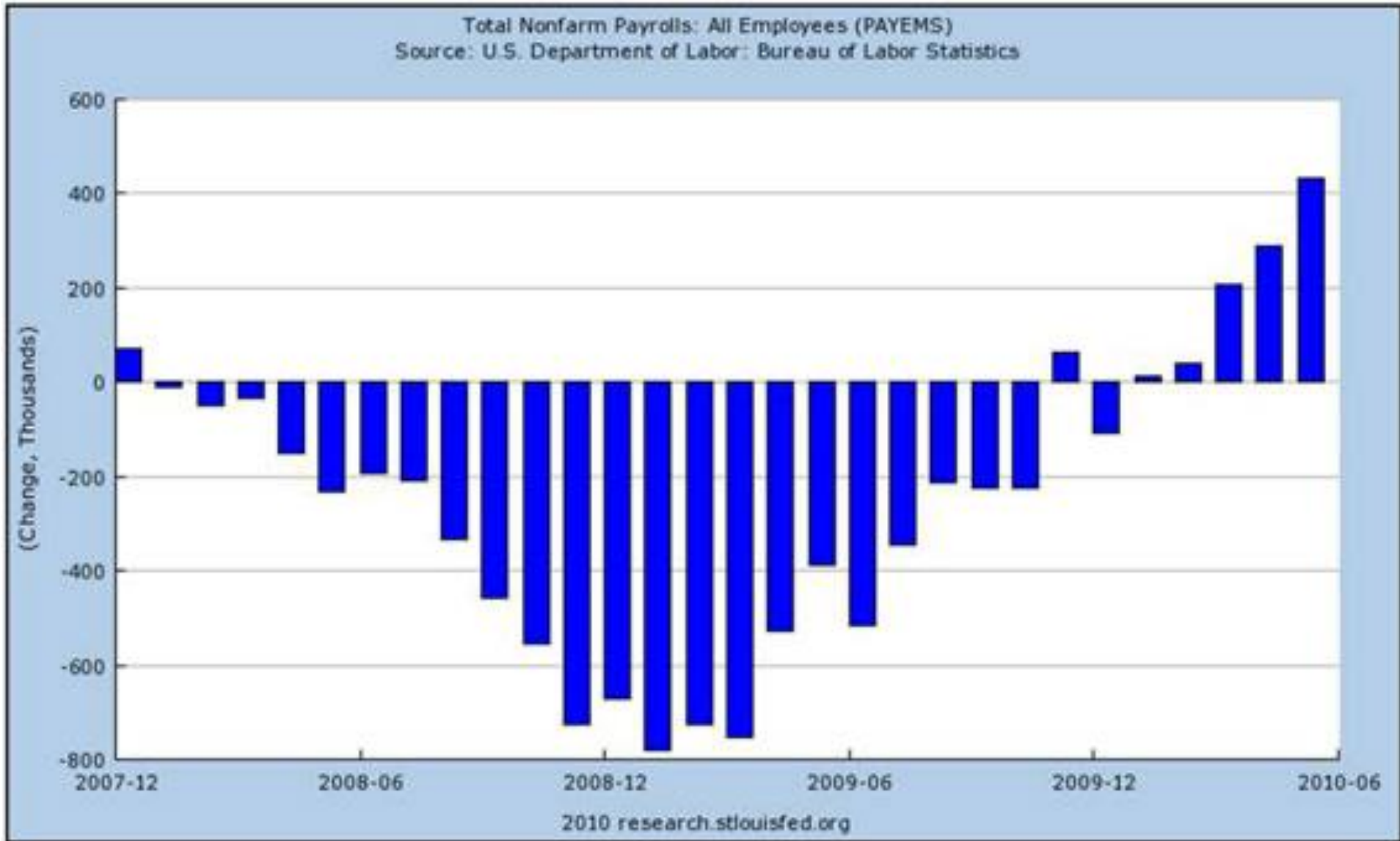
FOX NEWS
FOX NEWS
.COM

SOURCE: BLS

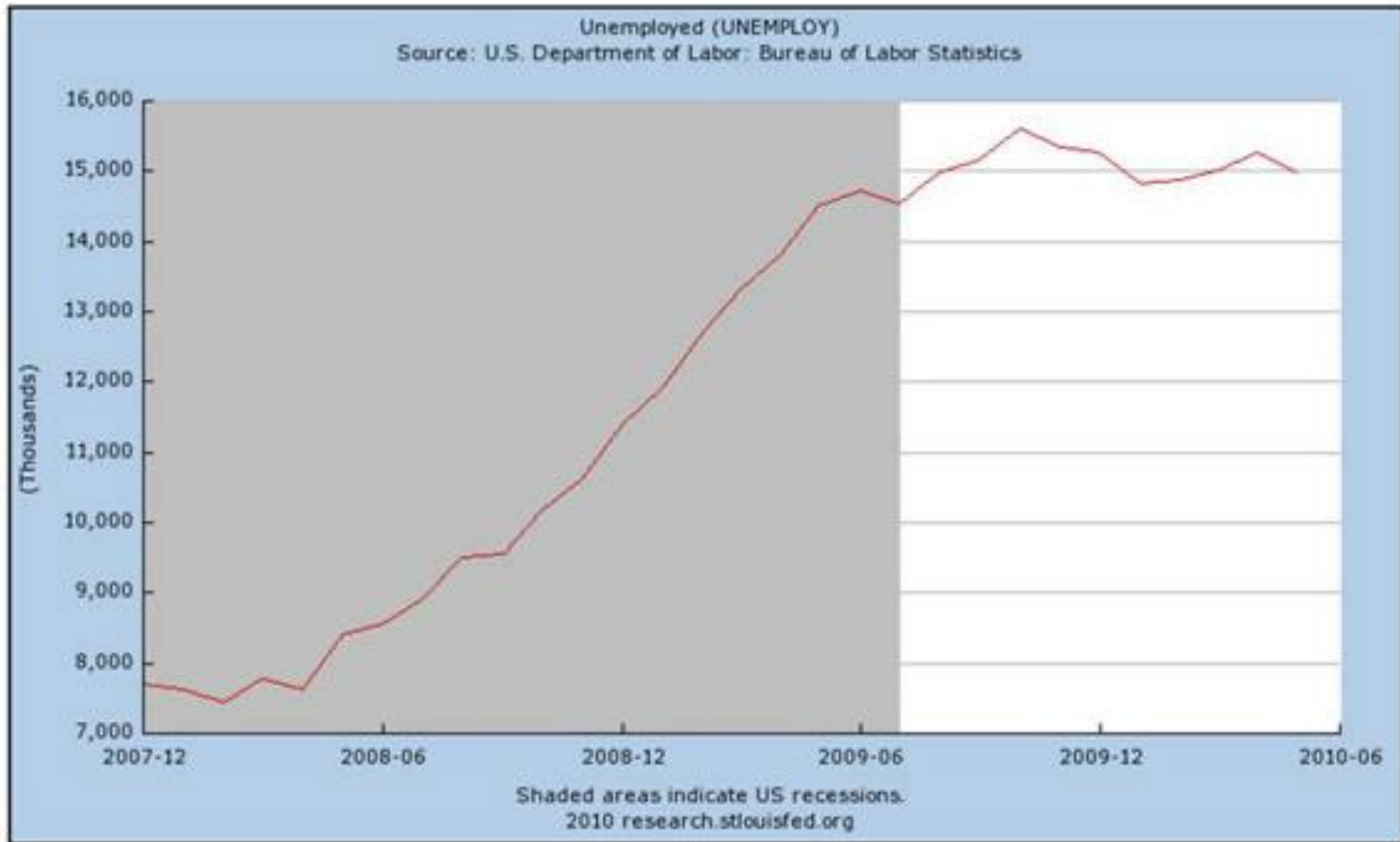
AMERICA'S
NEWSROOM

N FAIRFAX, VA... BYRD WAS ADMITTED TO THE HC

S&P ▼ 3.08



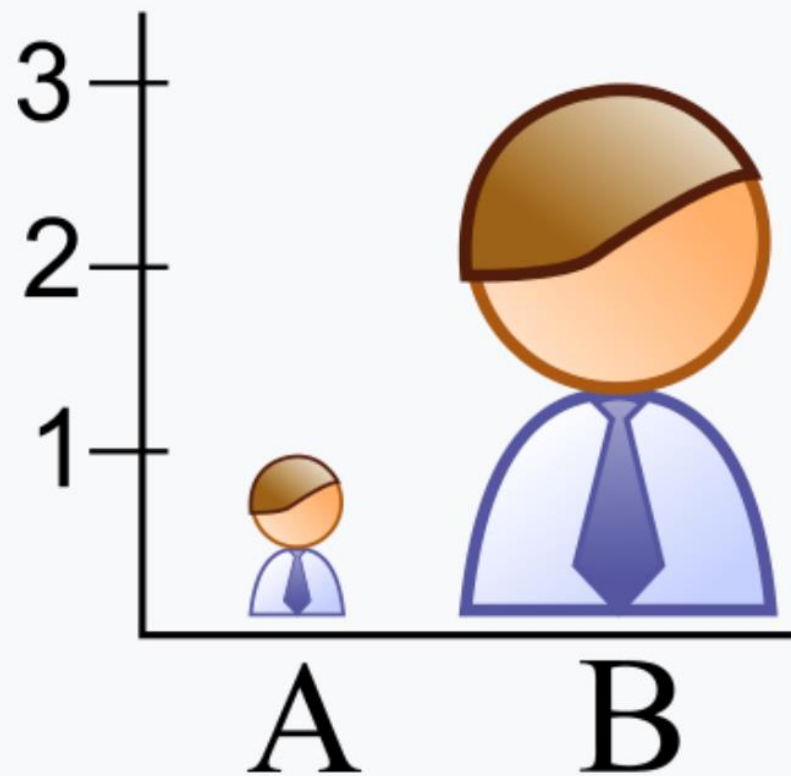




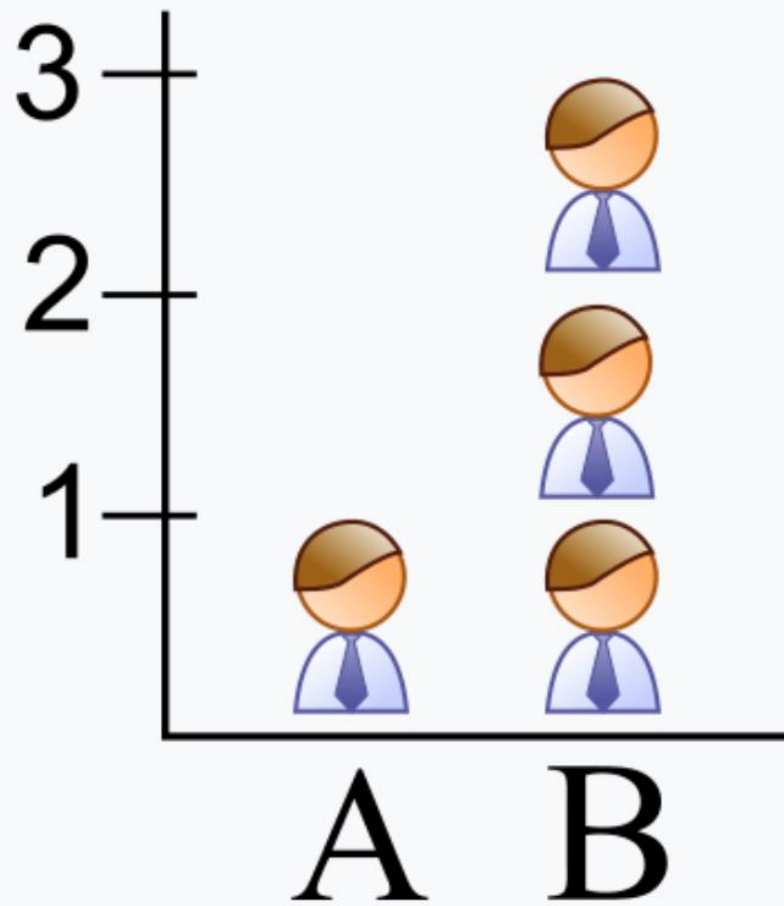
Common Ways that Visualizations Mislead

- **Fabrication**
- **Omission**
- **Framing**
 - Scaling (lineic vs areic vs volumic comparisons)
 - Truncating axes / omitting the baseline / axis ratios
 - Missing context / denominator problems
 - Going against convention

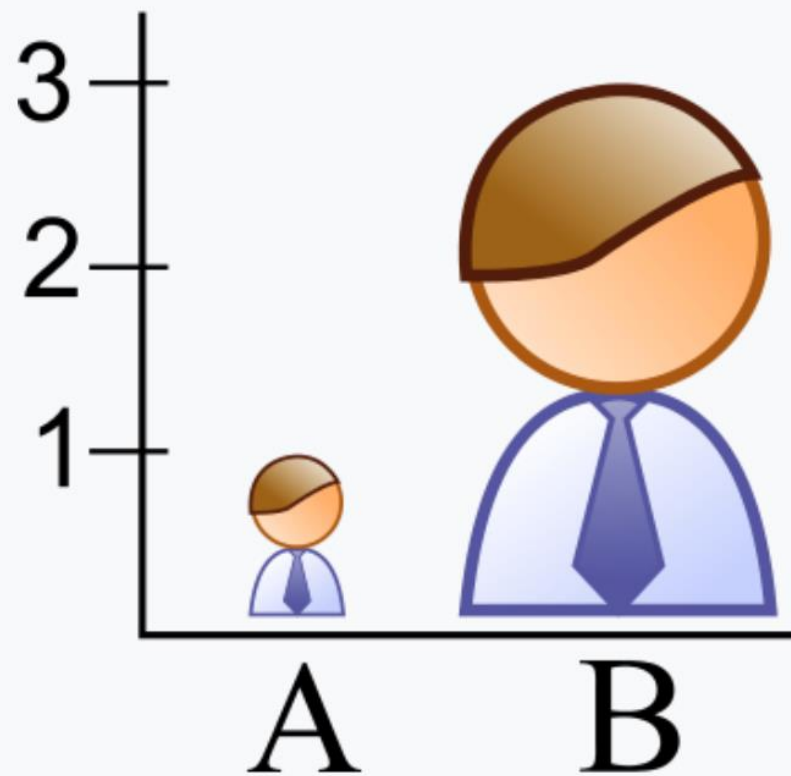
Improper scaling



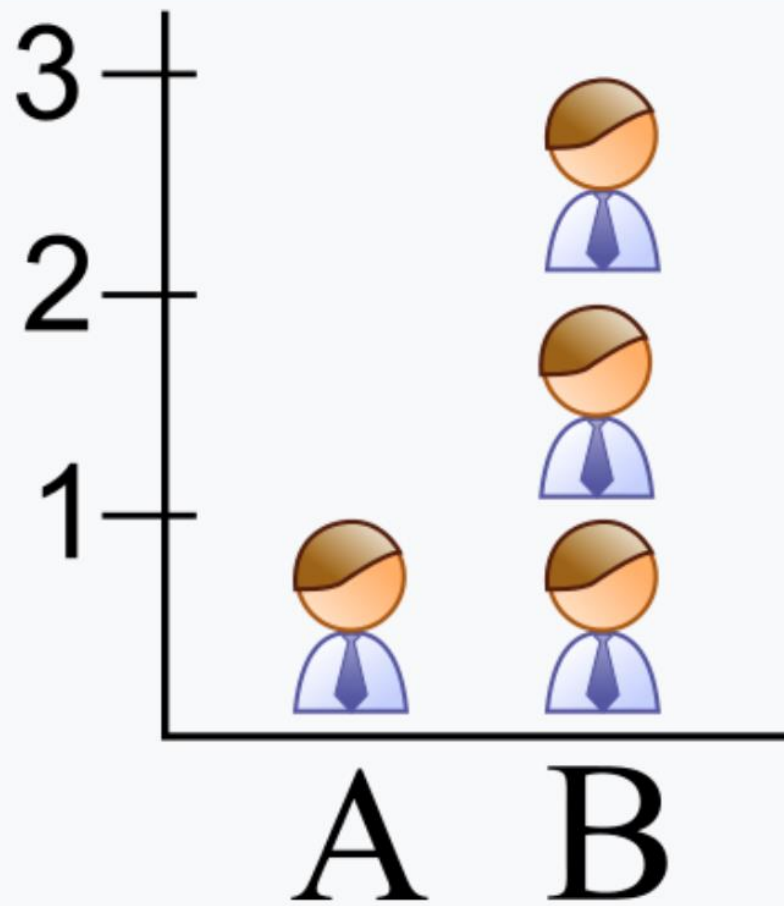
Regular



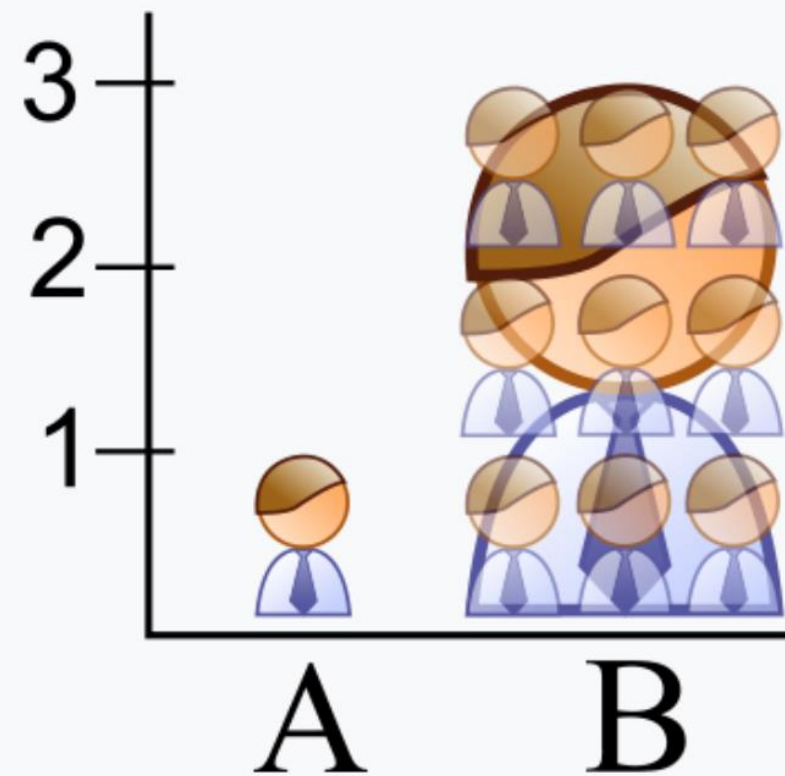
Improper scaling



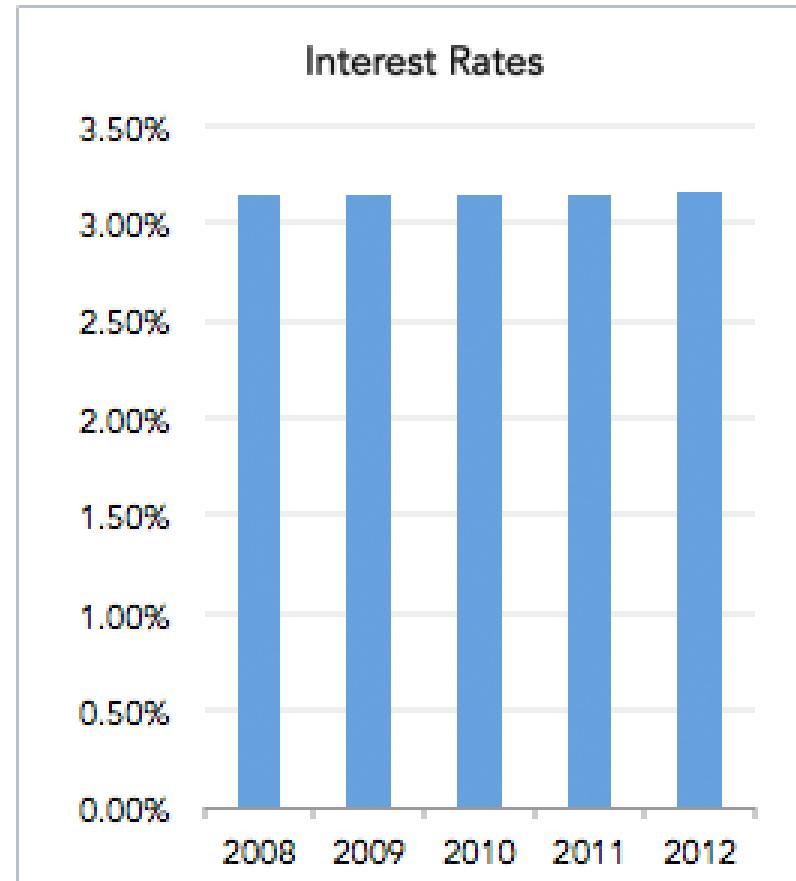
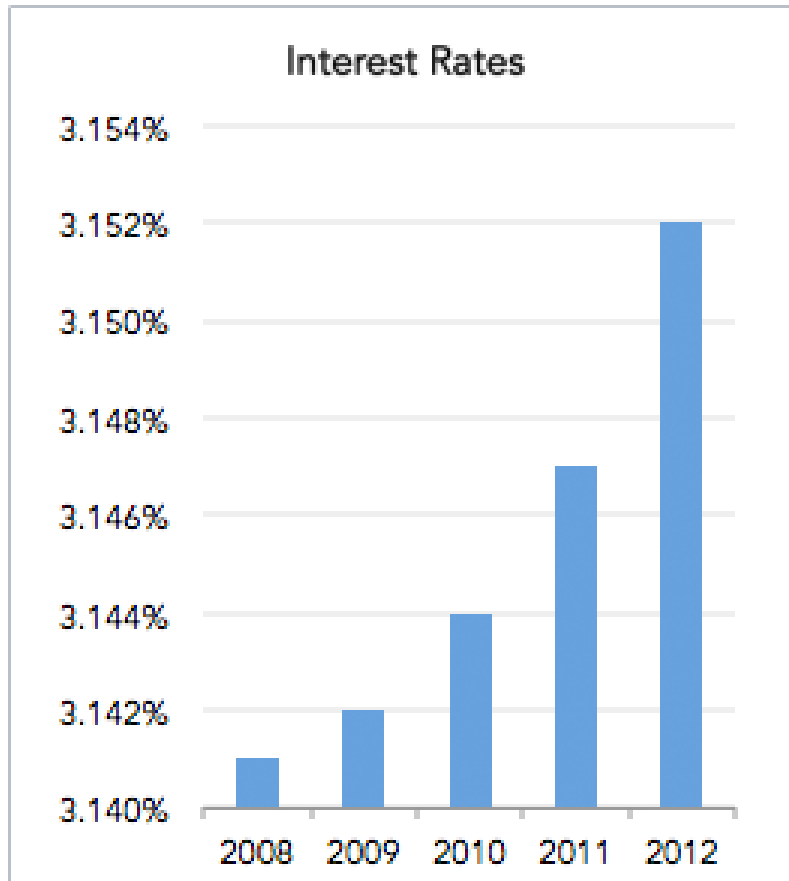
Regular



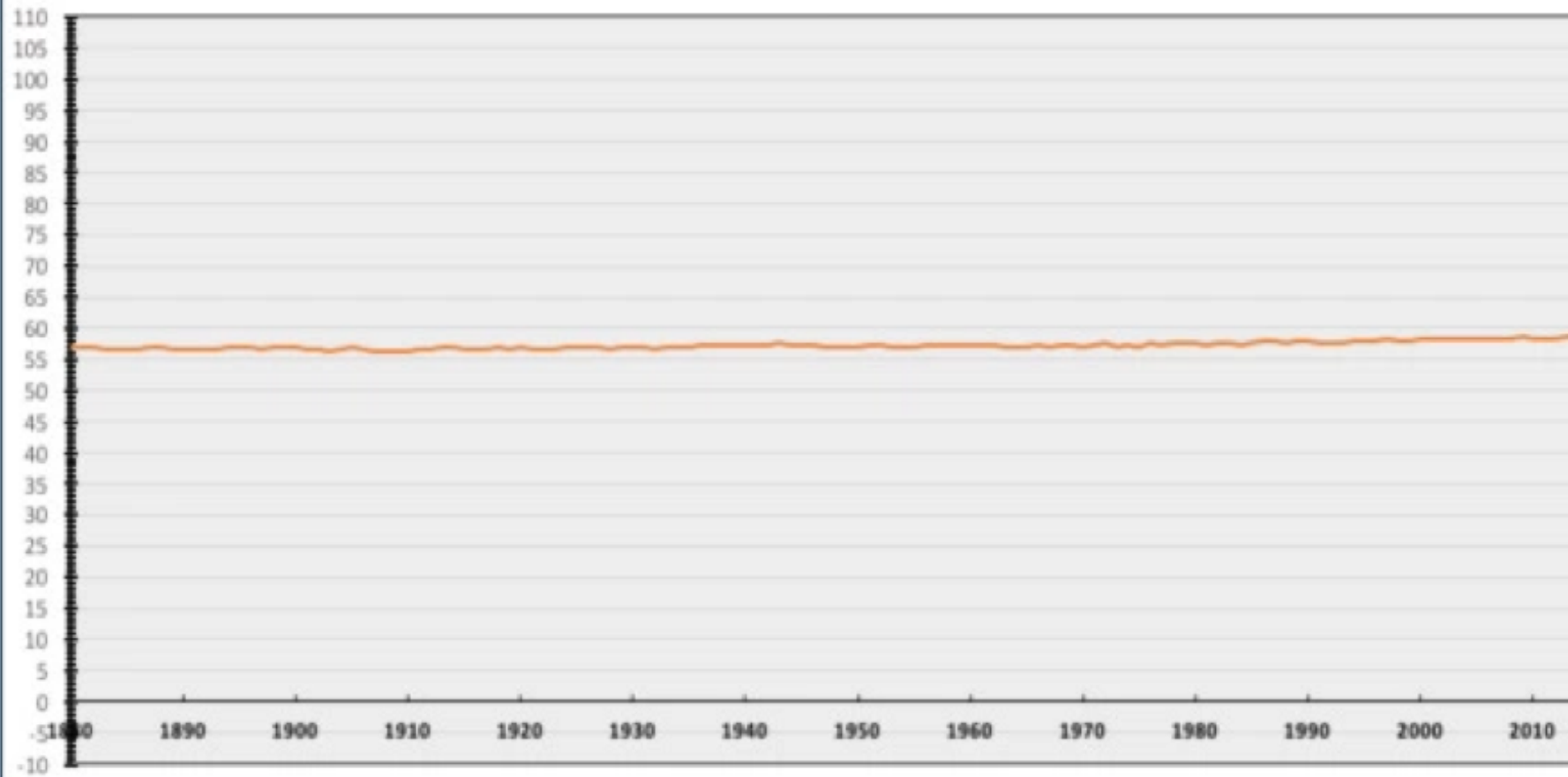
Comparison



Same Data, Different Y-Axis

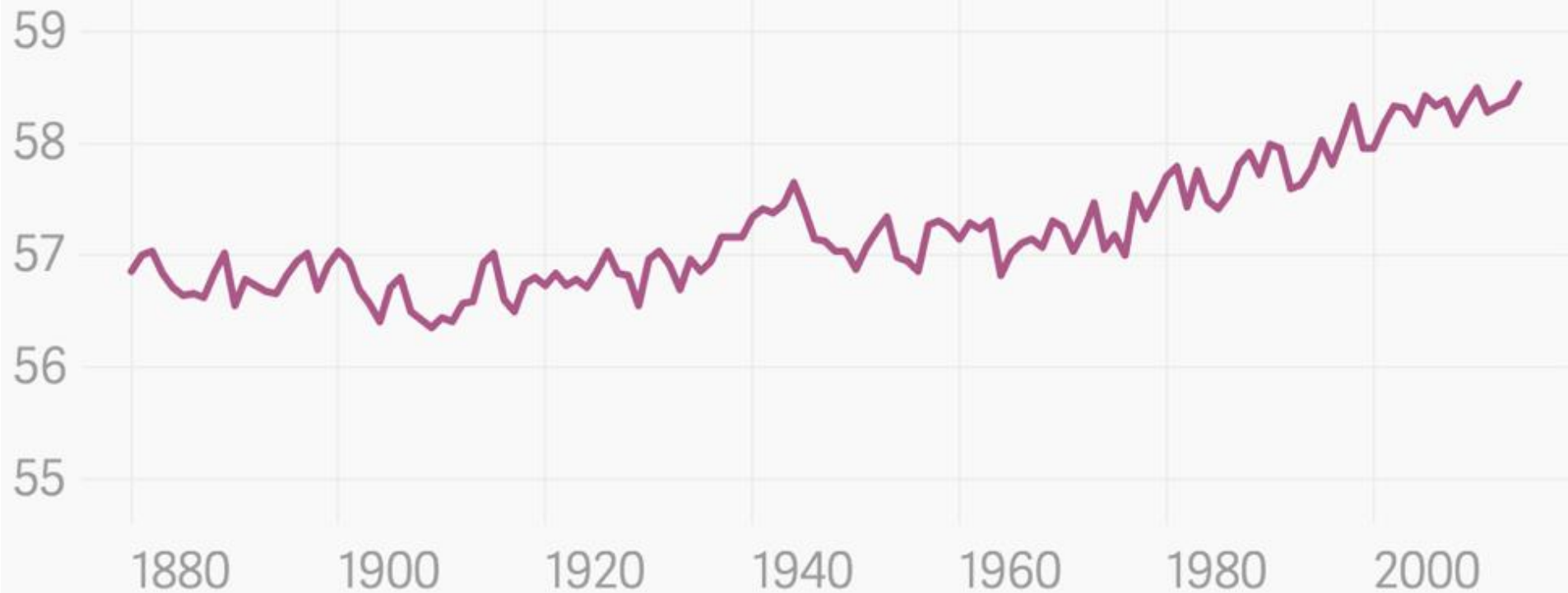


Average Annual Global Temperature in Fahrenheit
1880-2015



Average global temperature, 1880 to 2014

60° fahrenheit



△ T L △ S | Data: NASA

JOB LOSS BY QUARTER



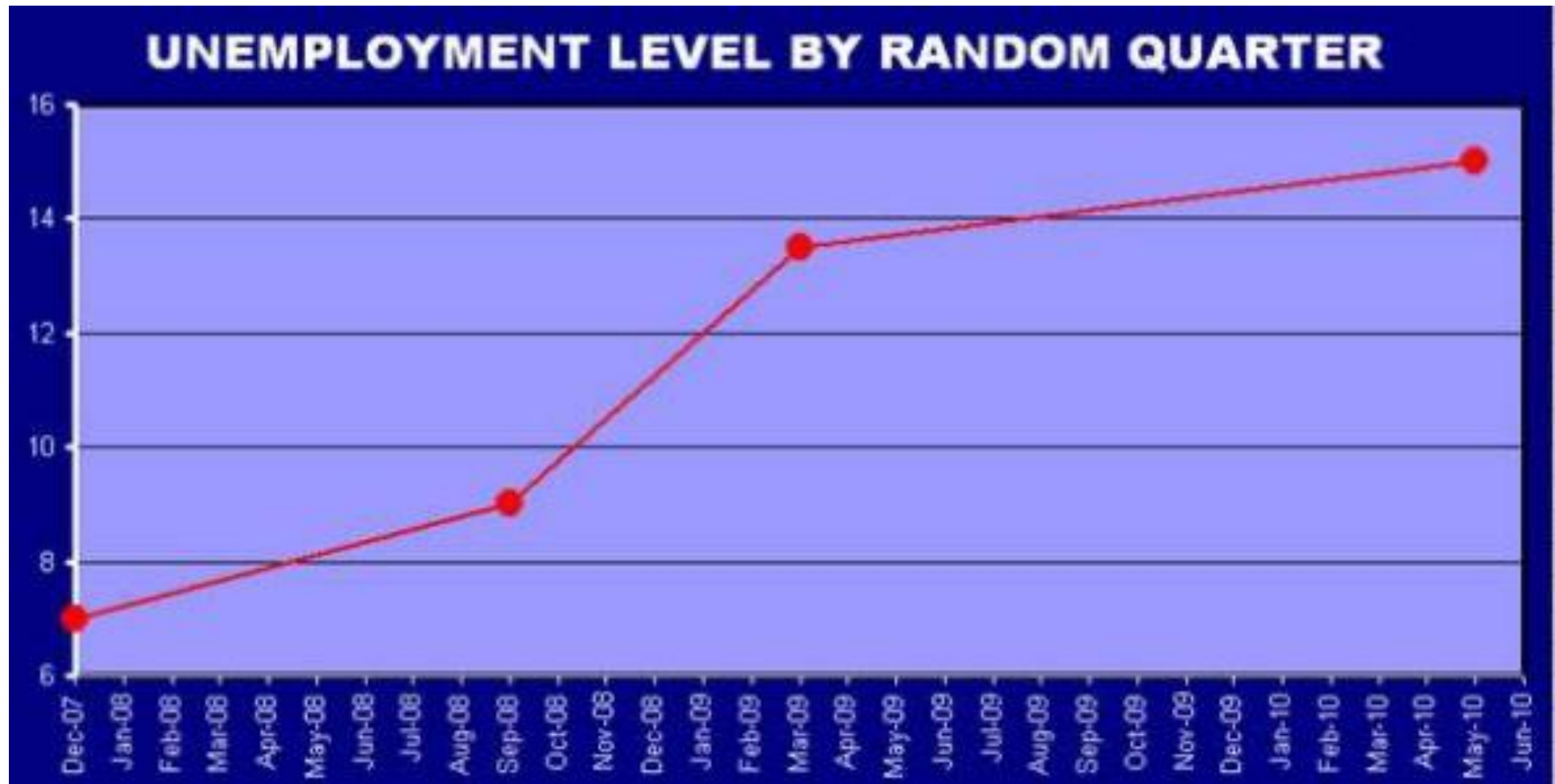
FOX NEWS
FOX NEWS
.COM

SOURCE: BLS

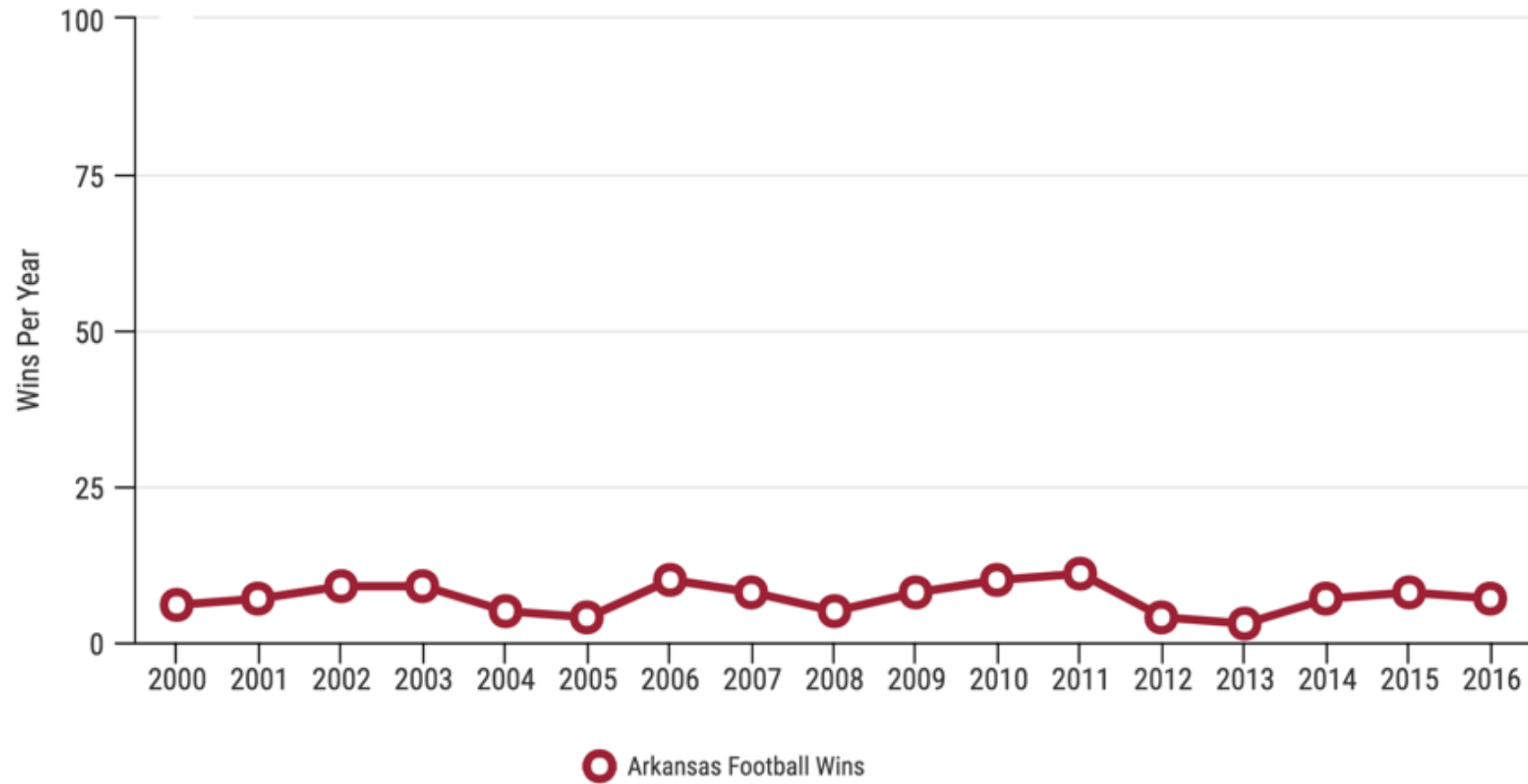
AMERICA'S
NEWSROOM

N FAIRFAX, VA... BYRD WAS ADMITTED TO THE HC

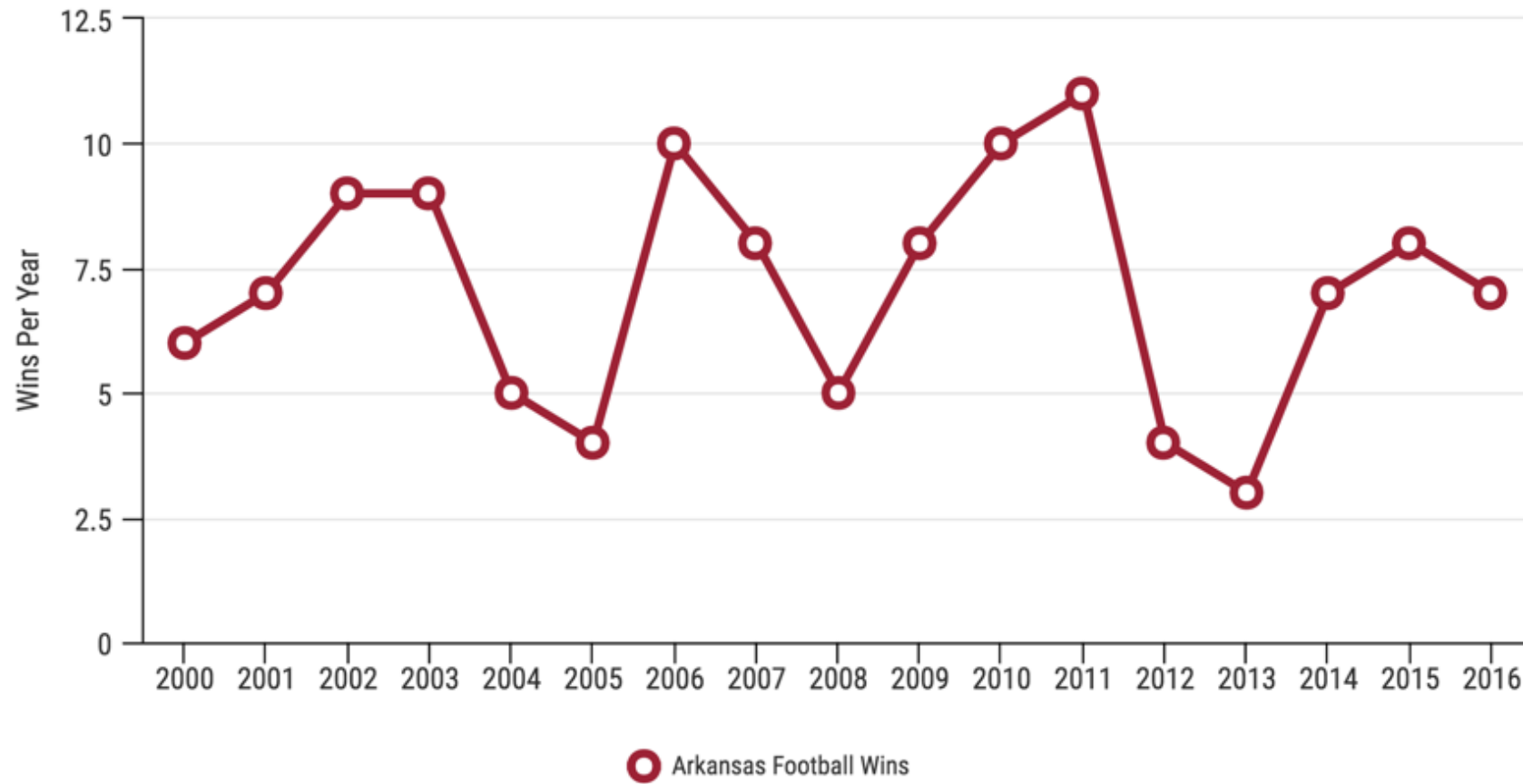
S&P ▼ 3.08



Bad Graph



Good Graph



Twitter Inc

NYSE: TWTR - May 24, 3:33 PM EDT

17.97 USD ↓0.17 (0.94%)

1 day

5 day

1 month

3 month

1 year

5 year

max



Open 18.10
High 18.34
Low 17.83

Mkt cap 12.93B
P/E ratio -
Div yield -

Twitter Inc

NYSE: TWTR - May 24, 3:34 PM EDT

17.98 USD ↓0.16 (0.91%)

1 day

5 day

1 month

3 month

1 year

5 year

max

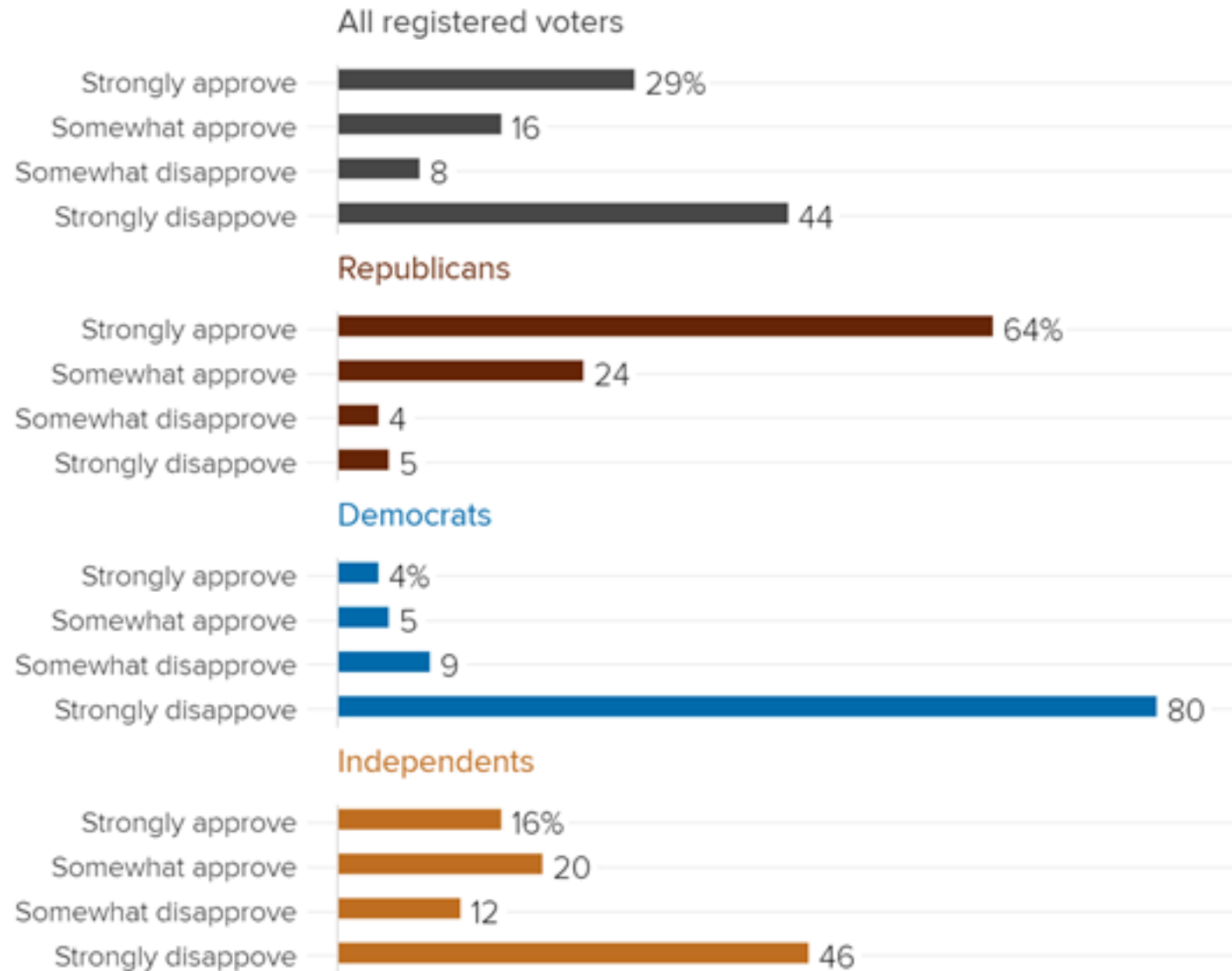


Open 18.10
High 18.34
Low 17.83

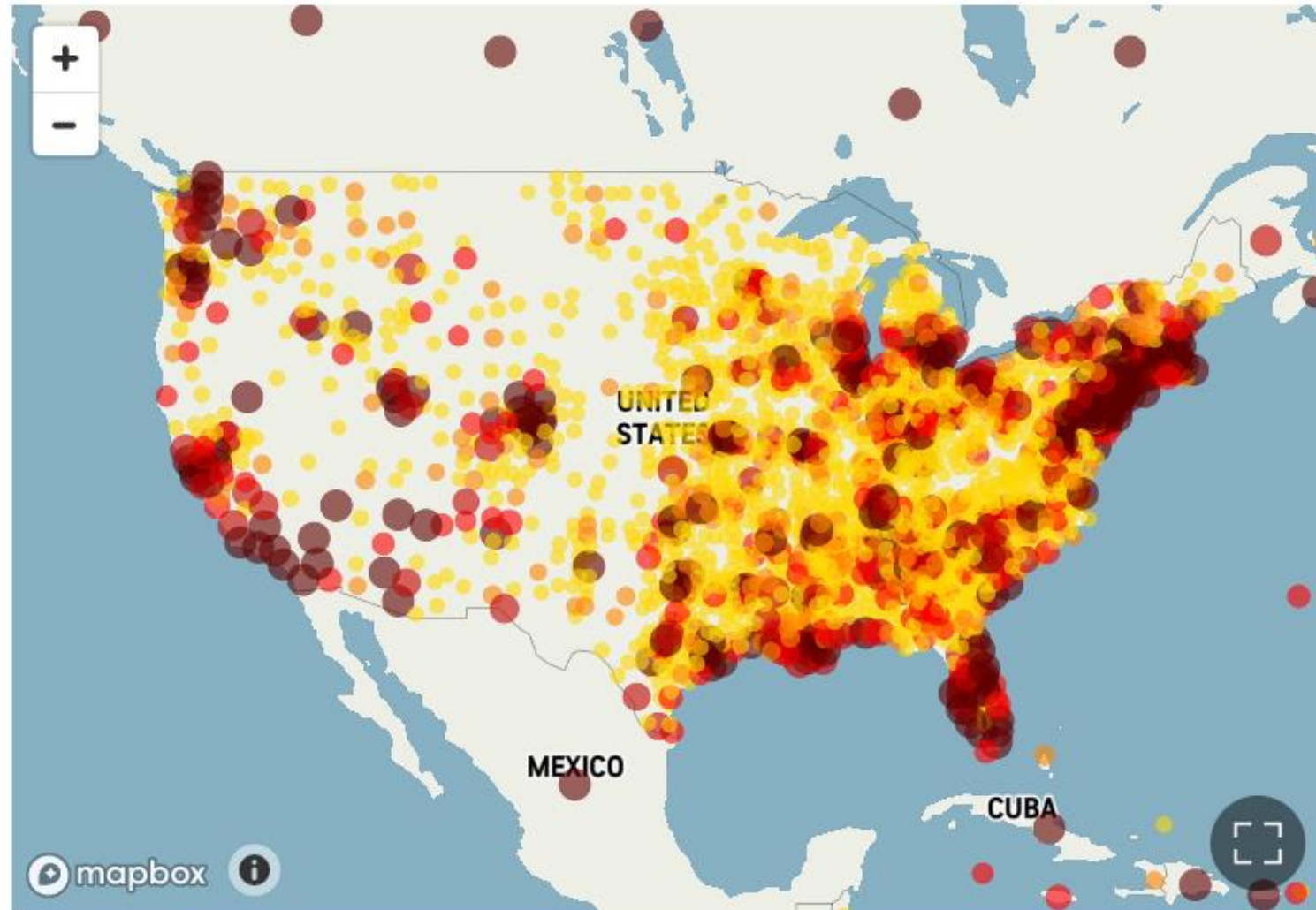
Mkt cap 12.93B
P/E ratio -
Div yield -



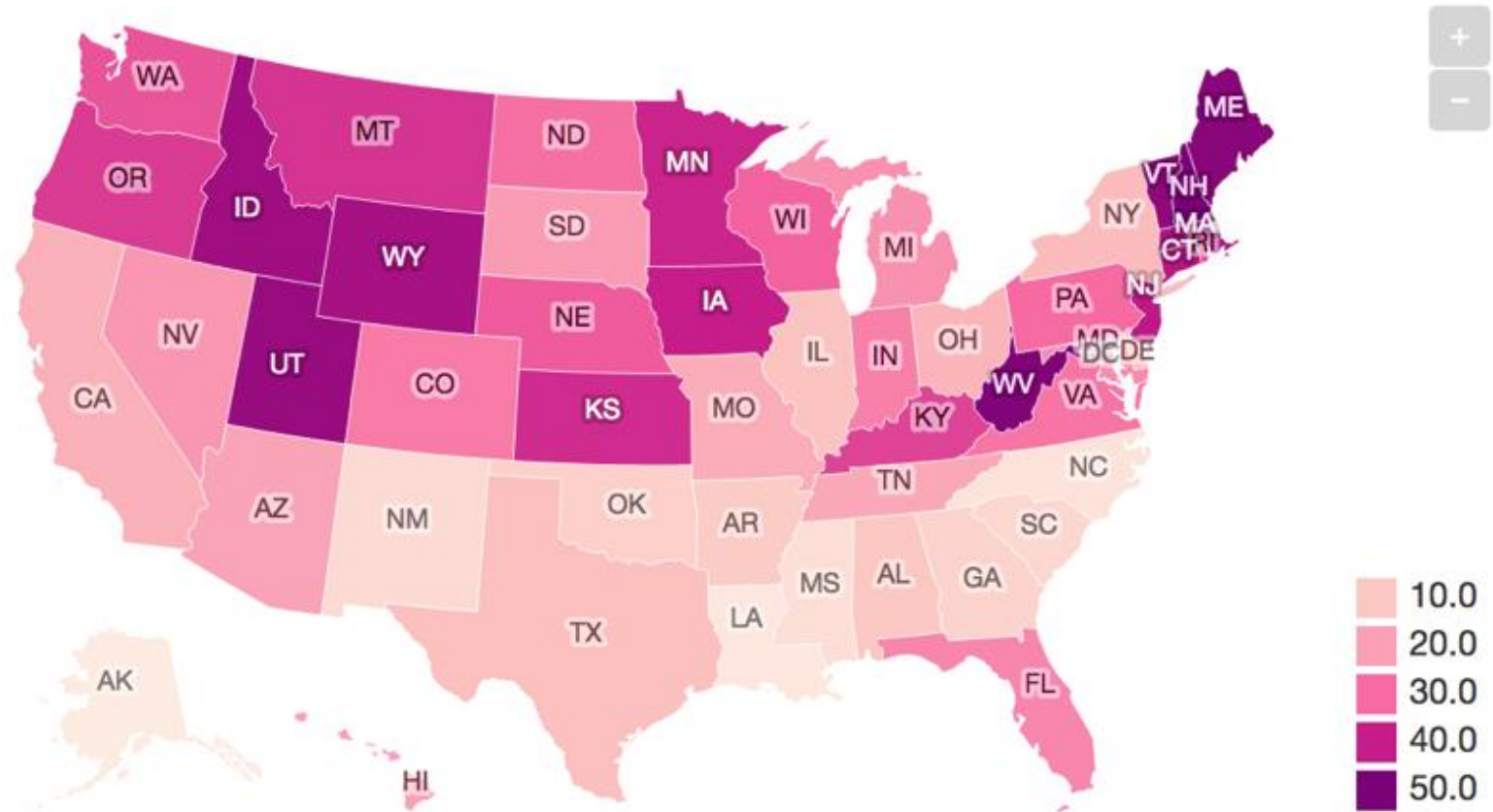
Strength of Trump approval/disapproval by party



CONFIRMED CASES BY REGION



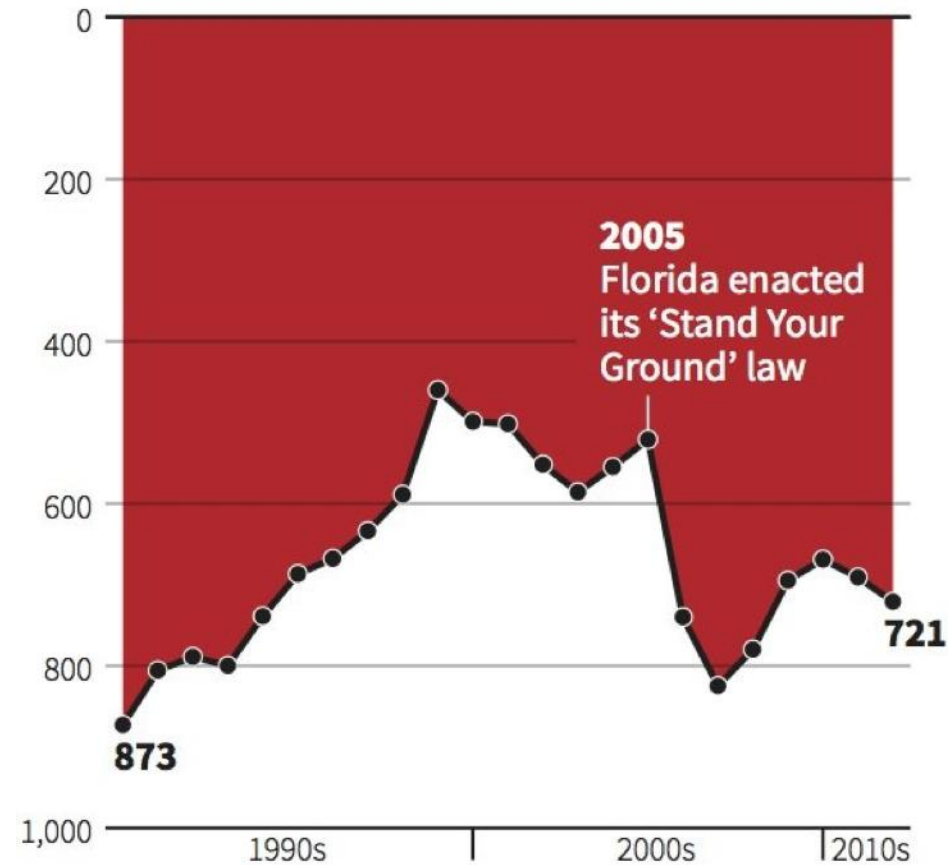
Which states have the most STIs?



[Get the data](#)

Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

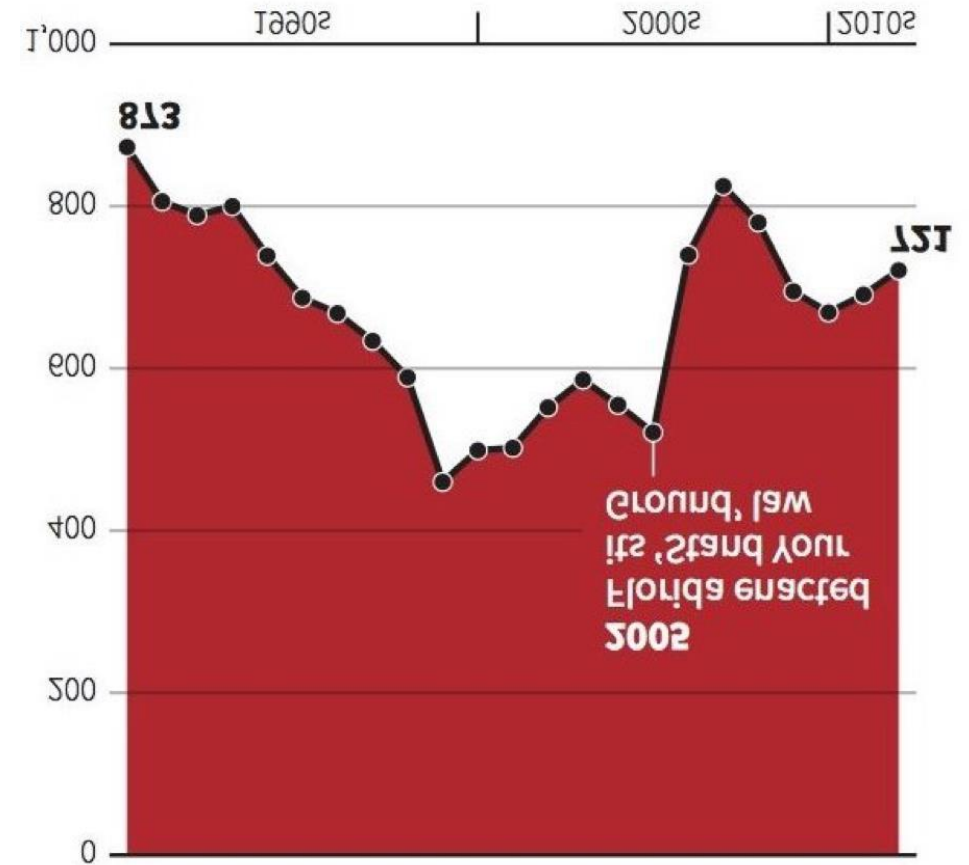
C. Chan 16/02/2014

REUTERS

C. Chan 16/02/2014

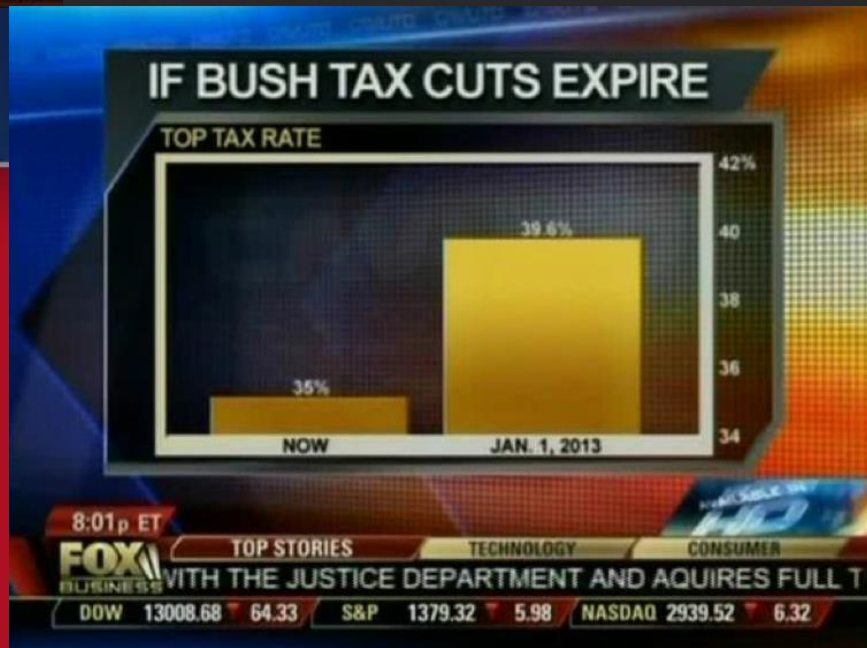
REUTERS

Source: Florida Department of Law Enforcement



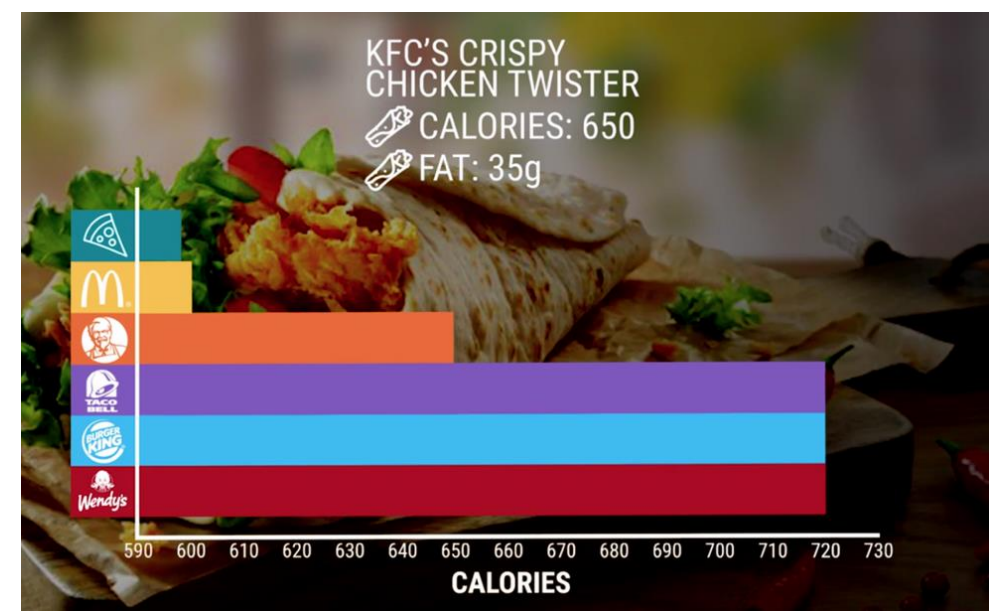
Number of murders committed using firearms

Gun deaths in Florida





Appellate Judgeships Confirmed During First Congressional Term. Ronald Reagan, 19; George Bush, 18; Bill Clinton, 18; George W. Bush, 16; Barack Obama, 15; Donald Trump, 24. Illustration by Tracy Ma



How to avoid misleading visualizations

- Be **conscious** of your **choices** (hopefully this class helps with that)
- Have **conversations** with your team about how the visualization is explaining or persuading
- Run visualizations by **non-specialists and non-insiders**:
 - What is their take-away?
 - Are you unintentionally misleading?
- Run a visualization by someone for only **5 seconds**:
 - What is their take-away?
 - Are you unintentionally misleading?

Manipulation of Images/Photographs

WHAT IS IMAGE MANIPULATION?

Cropping	Brightness/contrast adjustment	Selective enhancement	Colour adjustment	Cloning
Cropping that changes the context of the final image	Brightness/contrast adjustment that fades or vanishes part(s) of the image	Selective enhancement of the image with the intent to support the proposed hypothesis	Colour adjustment that distorts the observation and the conclusion from the image	Cloning an object into an image in which it did not appear previously

WHAT ARE THE IMPLICATIONS?

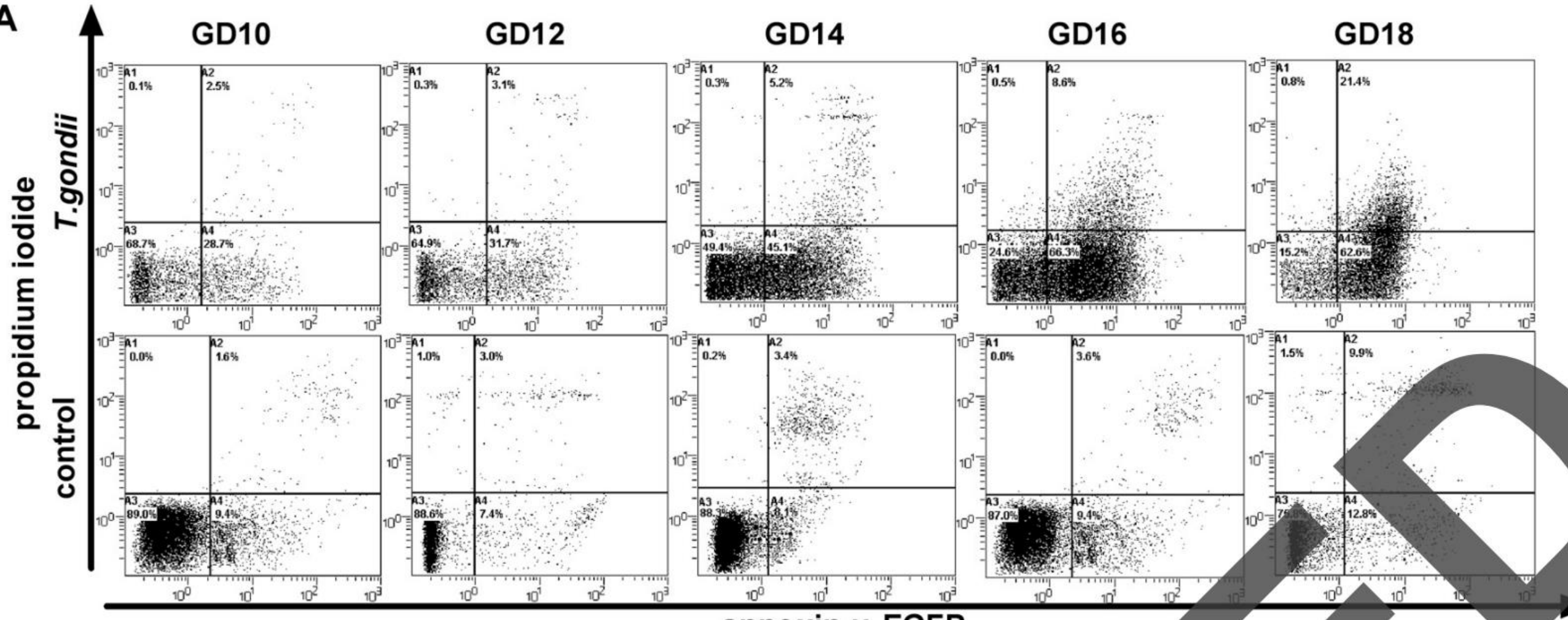
Affects research integrity. Considered a misconduct	Impacts reproducibility of the outcome	May result in paper retraction, damaged reputation and career, etc.

enago academy
Learn. Share. Discuss. Publish.
Source: <https://on.khs.gov/education/products/RandImages/practices/default.html>

- Image Integrity & Standards
 - <https://www.nature.com/nature-portfolio/editorial-policies/image-integrity>
- Specify equipment and acquisition software
- Deposit raw images with submission
- Processing software should be named, and manipulations should be enumerated

3.8% of Publications have Problematic Figures

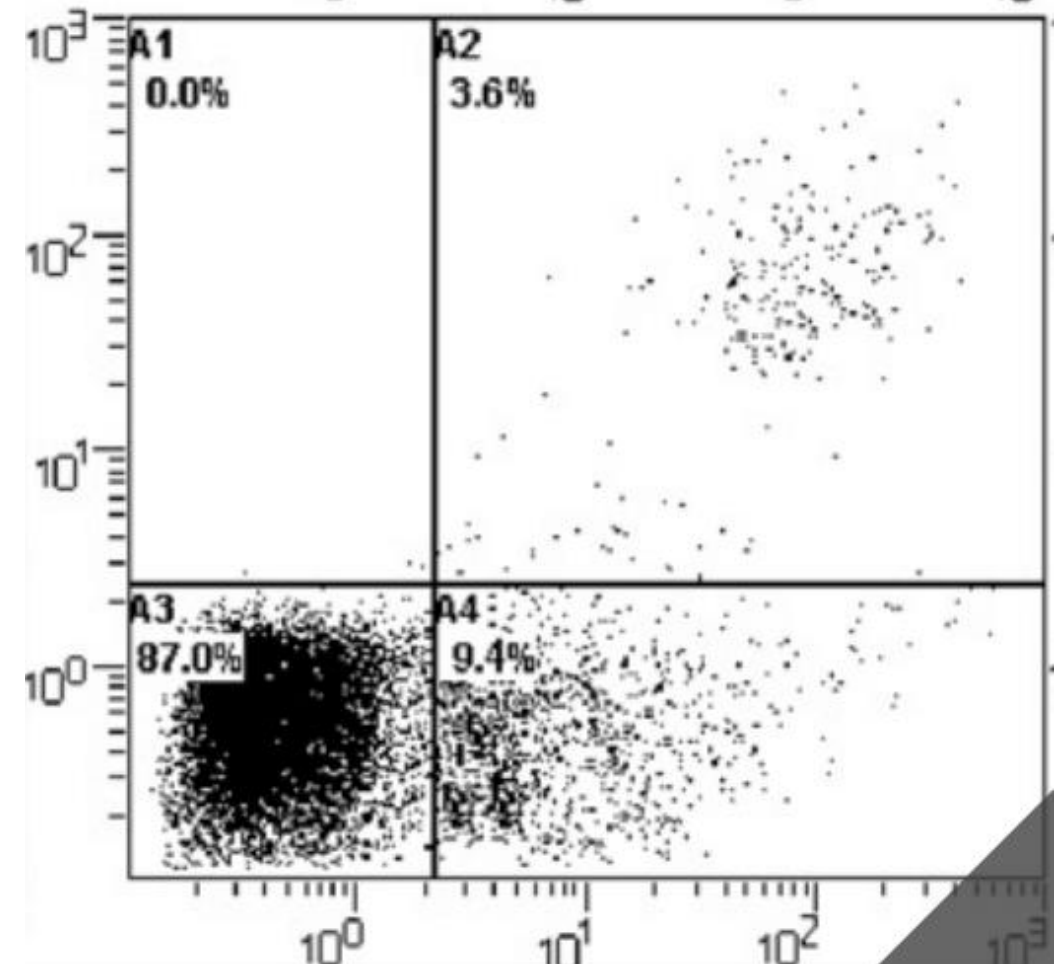
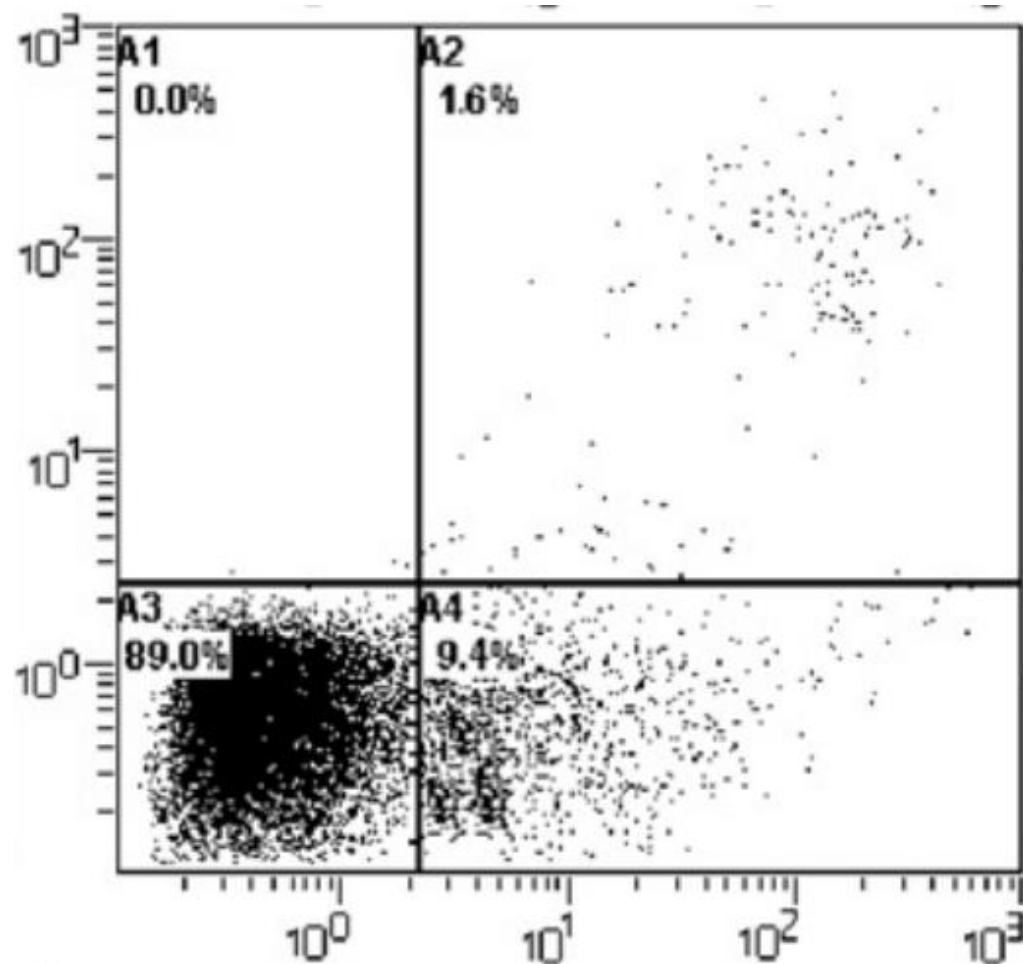
Bik *et al. mBio*. 2016



Xu *et al. Infect Immun*. 2012 (Retracted)

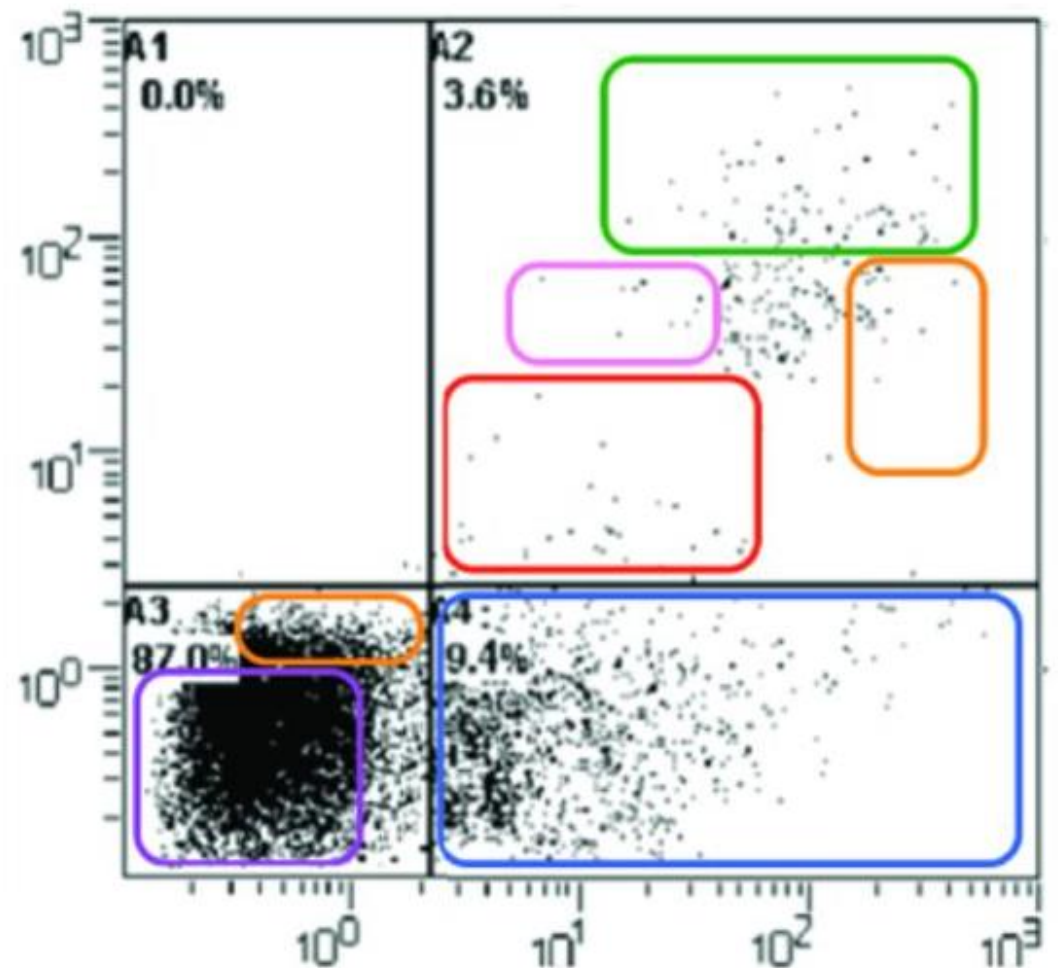
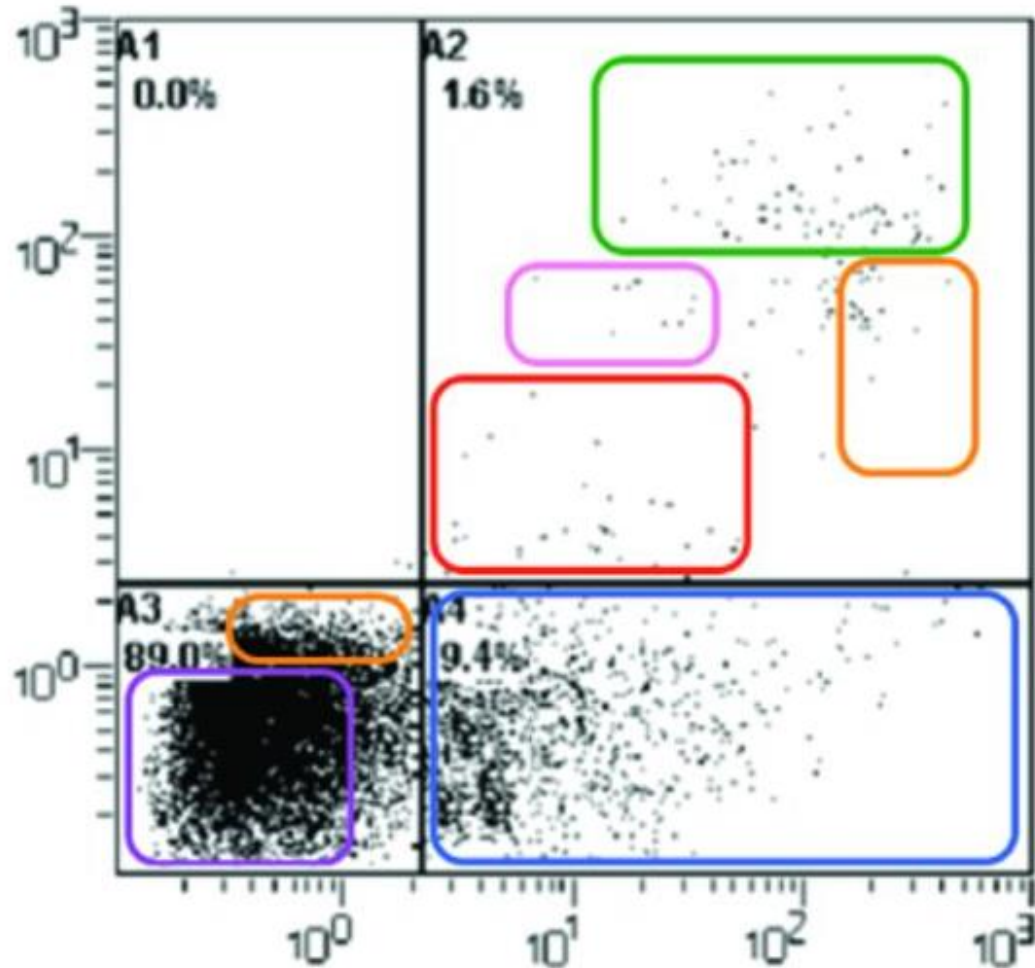
3.8% of Publications have Problematic Figures

Bik *et al. mBio*. 2016



Xu *et al. Infect Immun*. 2012 (Retracted)

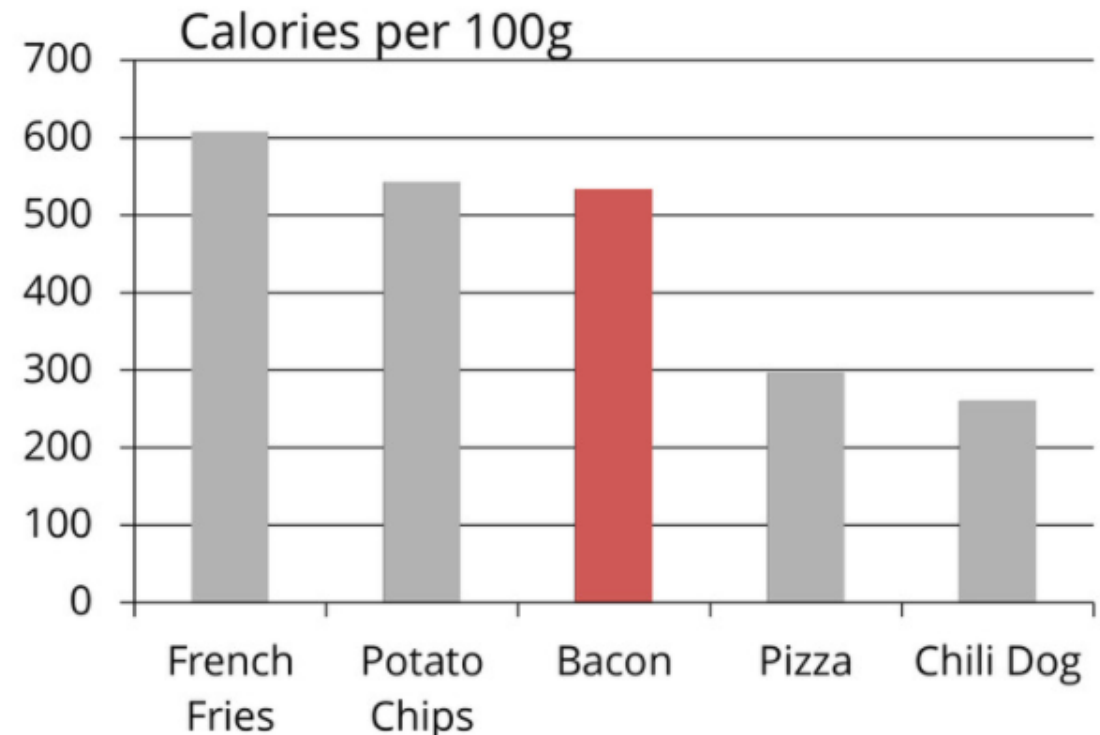
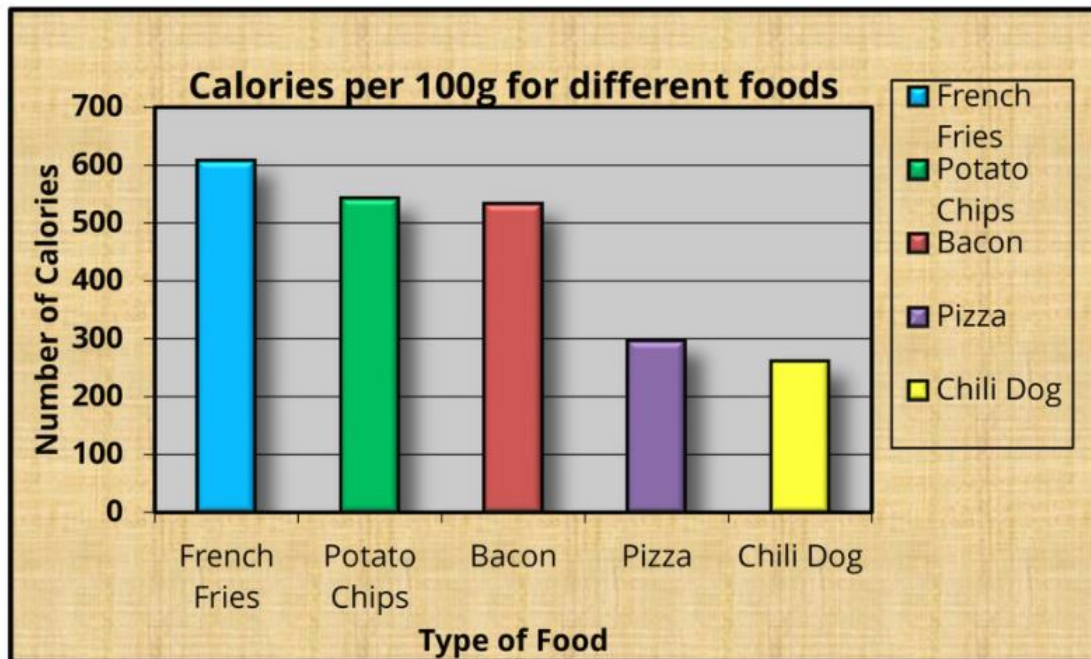
3.8% of Publications have Problematic Figures



Accessibility

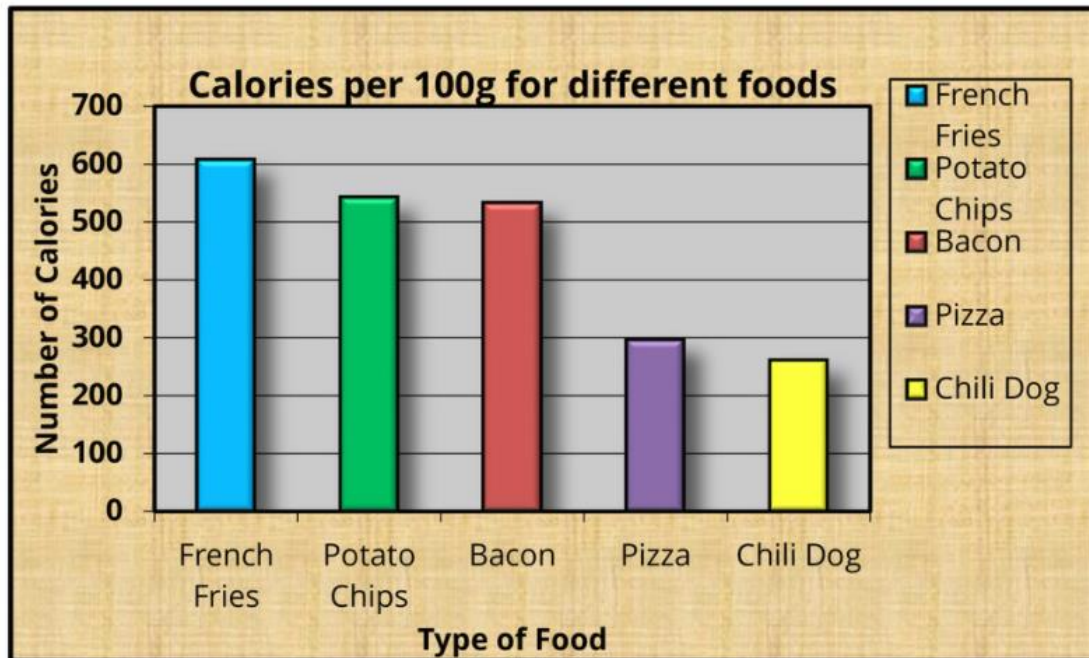
Accessibility

- In general, principles such as good data–ink ratios also inherently make images more accessible to individuals with visual impairments—both minimal and profound impairments

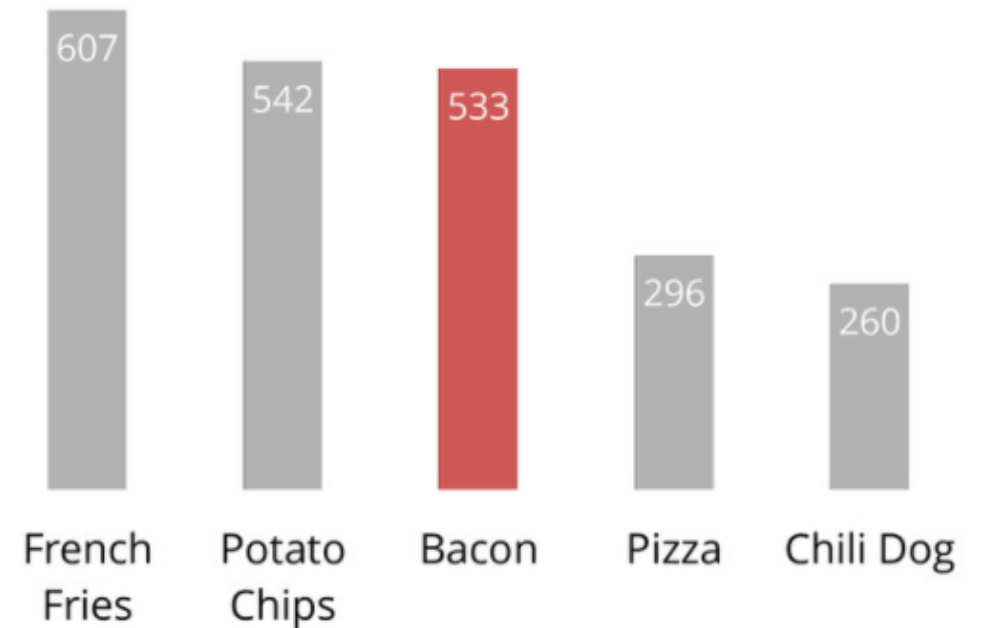


Accessibility

- But maybe not this:

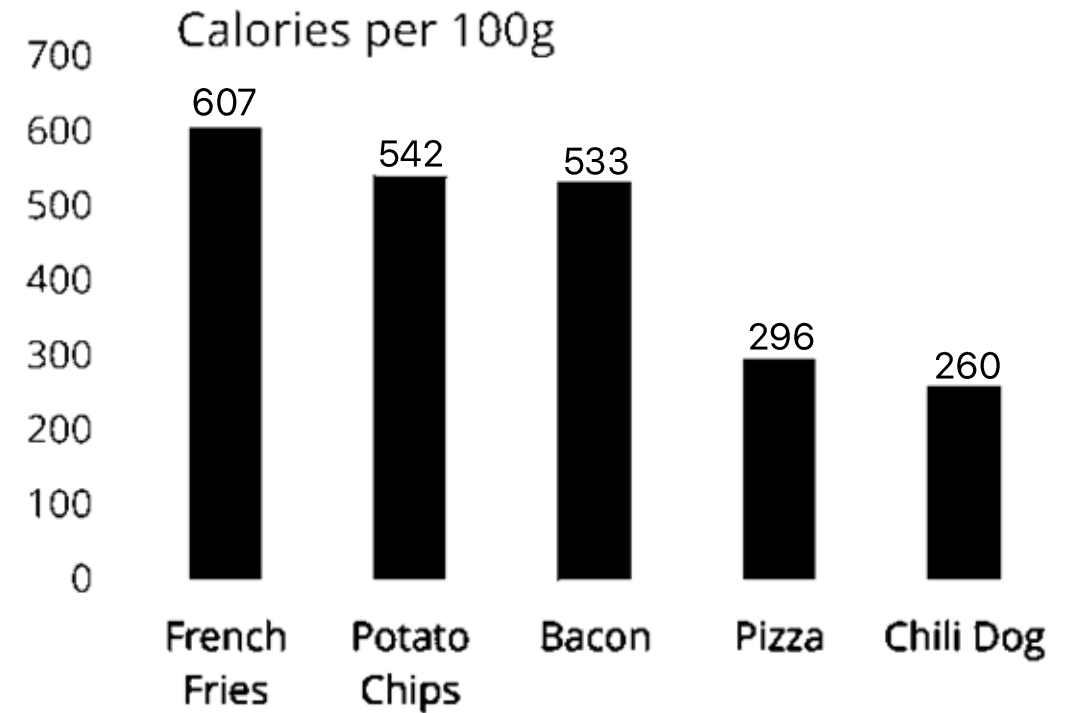
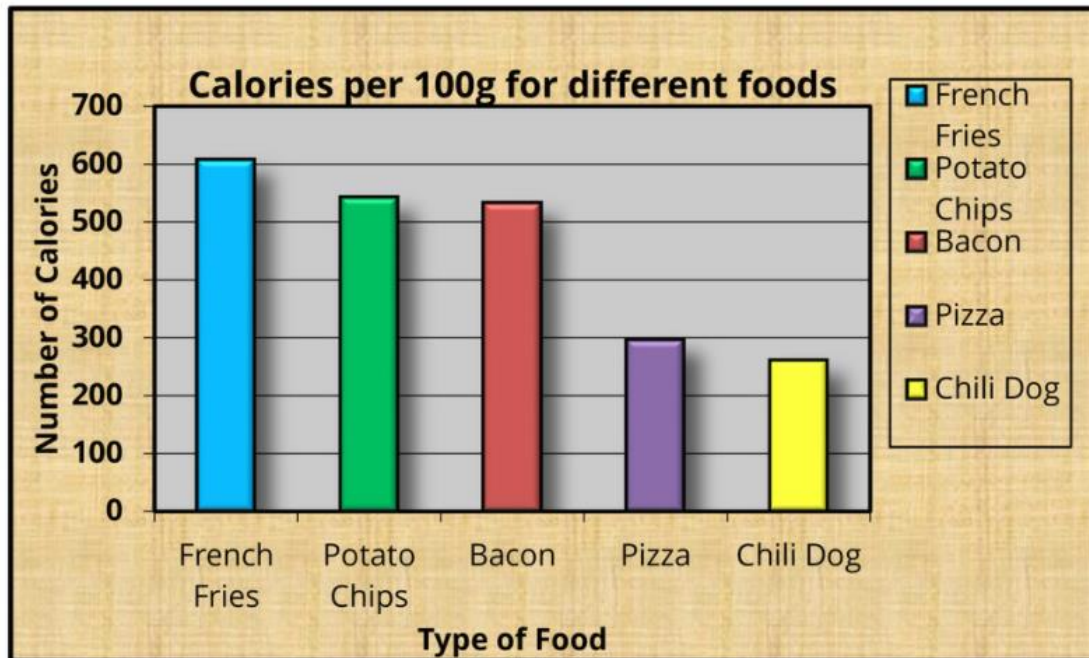


Calories per 100g



Accessibility

- Perhaps?



Accessibility

- Add **good alt text**
 - Example **guidelines** and **tools** for training: diagramcenter.org
- Use a **good title**
 - **Don't:** Risk of type 2 diabetes vs BMI
 - **Do:** Risk of type 2 diabetes is higher at higher BMI
- Figure captions (beyond the title) and alt text should complement each other. Captions often describe *how* to interpret the figure. Alt text describes the figure itself.

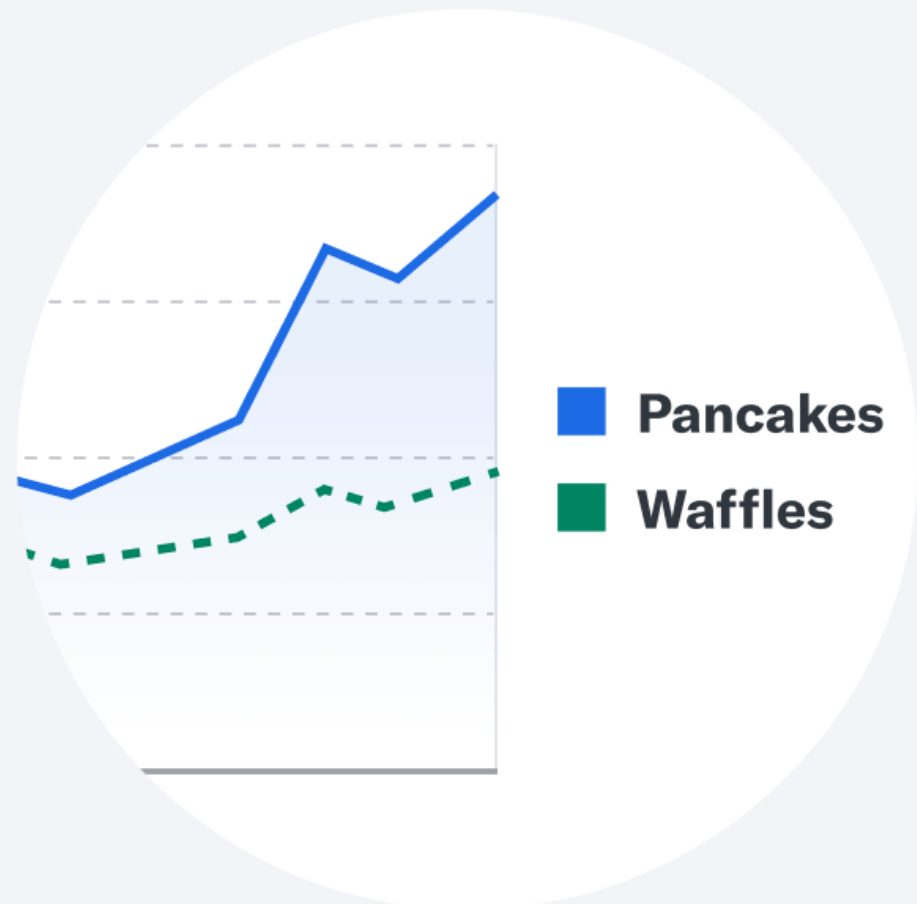
Accessibility

- Add **alt text**
- Use a **good title**
- **Label data directly**

DO



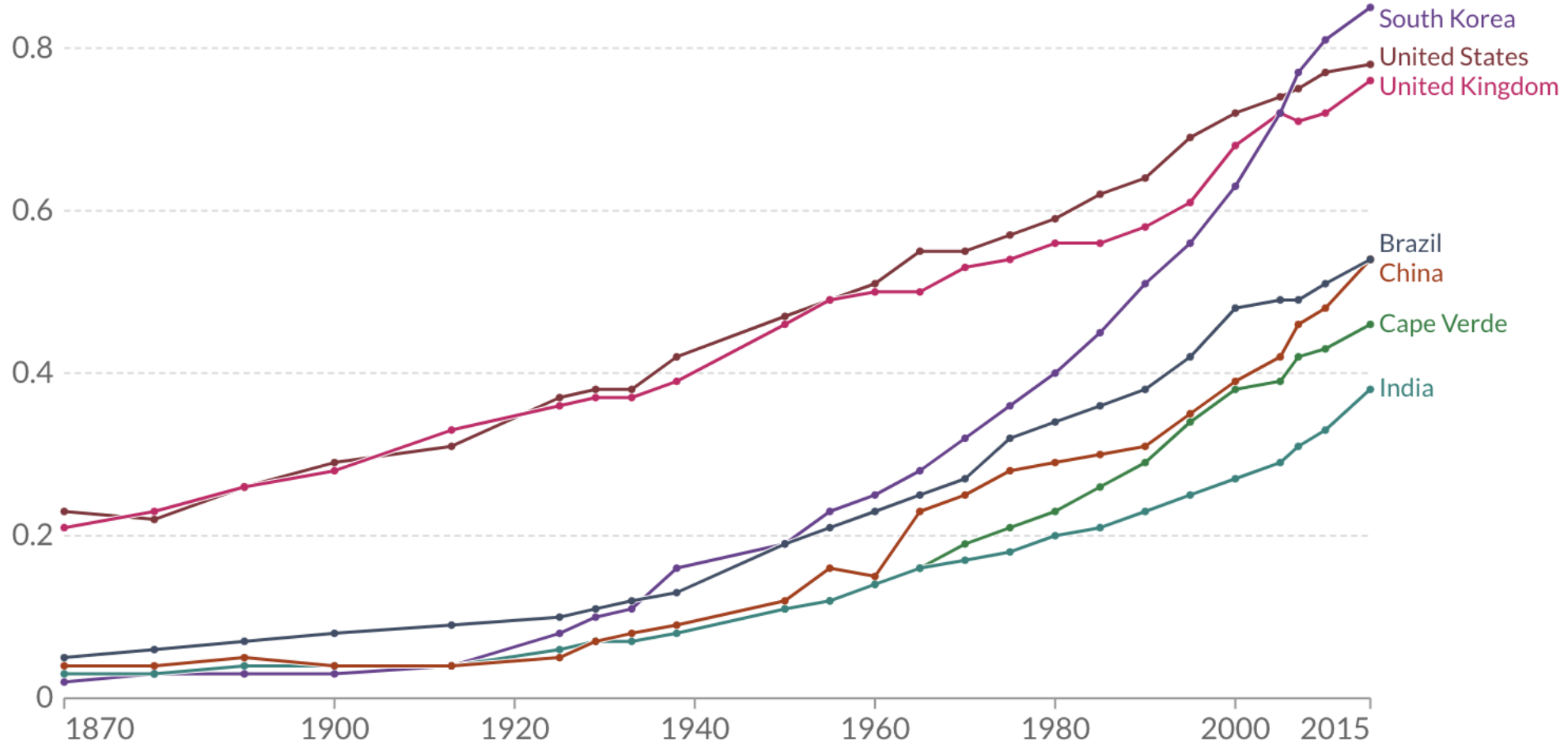
DON'T



Historical Index of Human Development, 1870 to 2015

The Historical Index of Human Development (HIHD) is a summary measure of average achievement in three key dimensions of human development: a long and healthy life, being knowledgeable and having a decent standard of living.

[+ Add country](#)



Source: Prados de la Escosura (2018)

OurWorldInData.org/human-development-index • CC BY



1870



2015

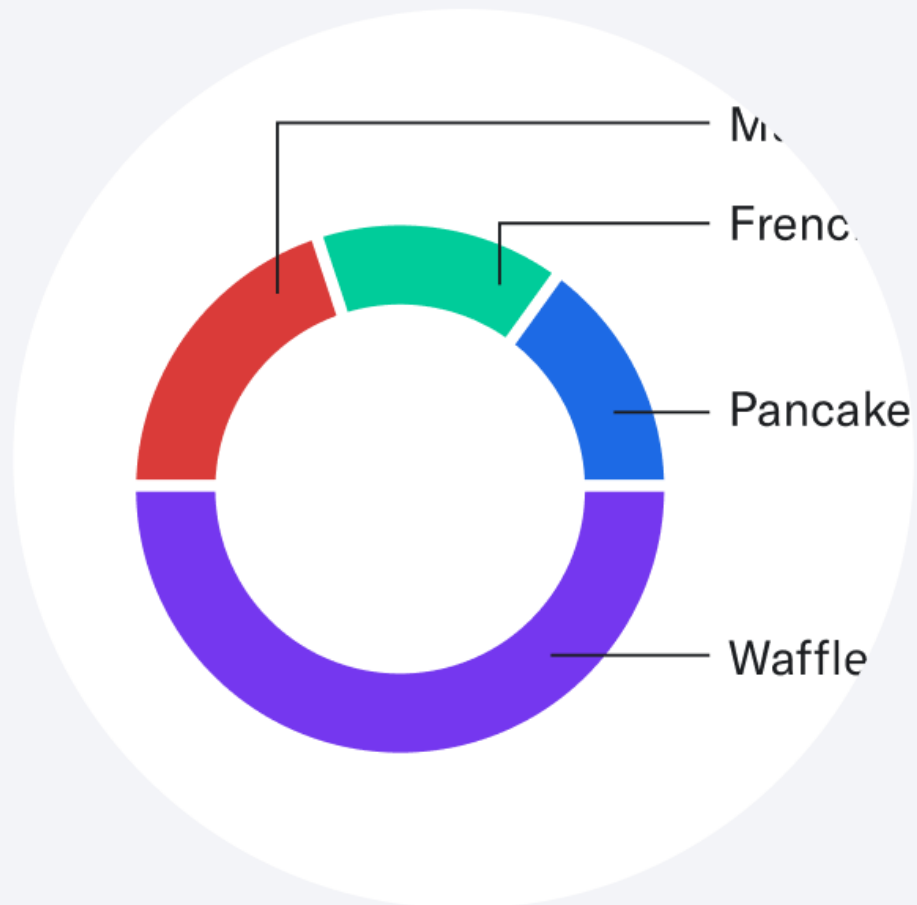
Accessibility

- Add **alt text**
- Use a **good title**
- **Label data directly**
- Employ **effective color contrast**
 - `colorsafe.co`
 - `colorbrewer2.org`
 - Colorblinding Chrome extension

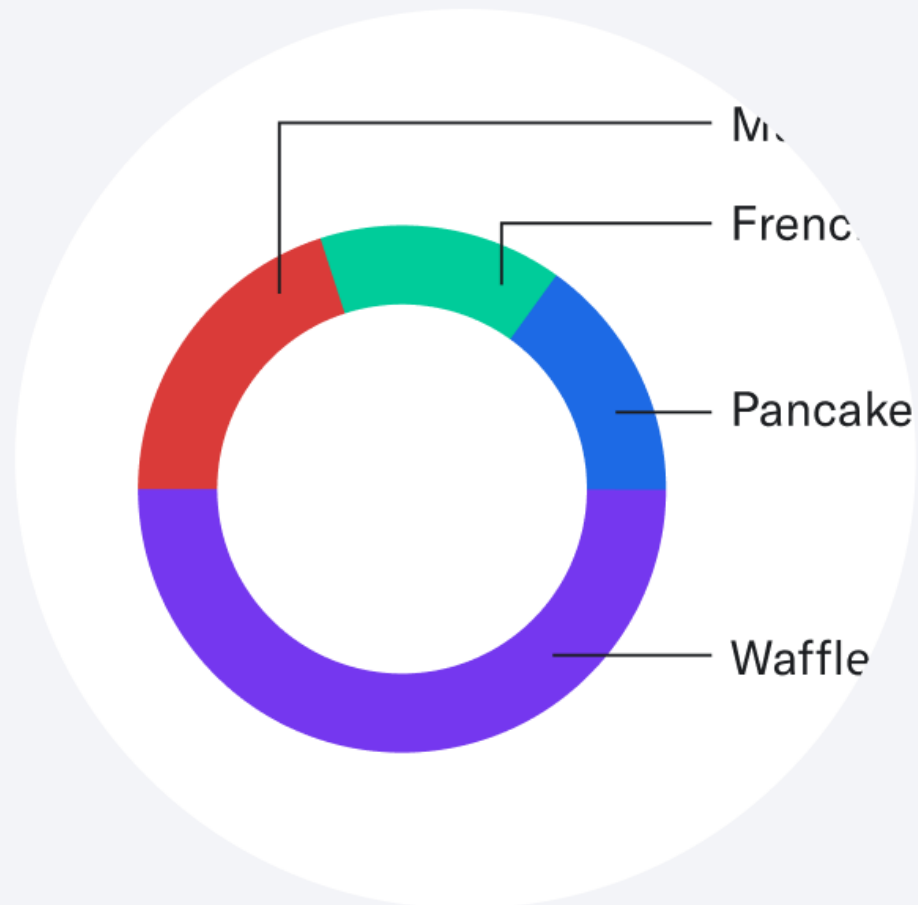
Accessibility

- Add **alt text**
- Use a **good title**
- **Label data directly**
- Employ **effective color contrast**
- Use **whitespace** effectively

DO

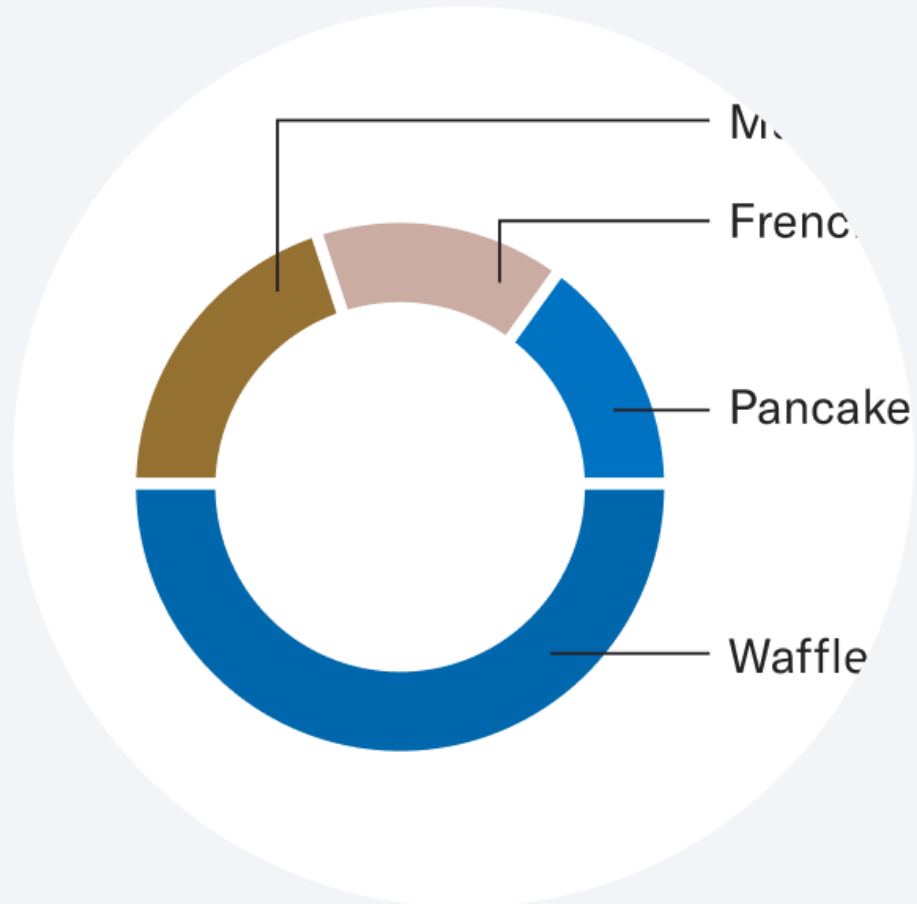


DON'T

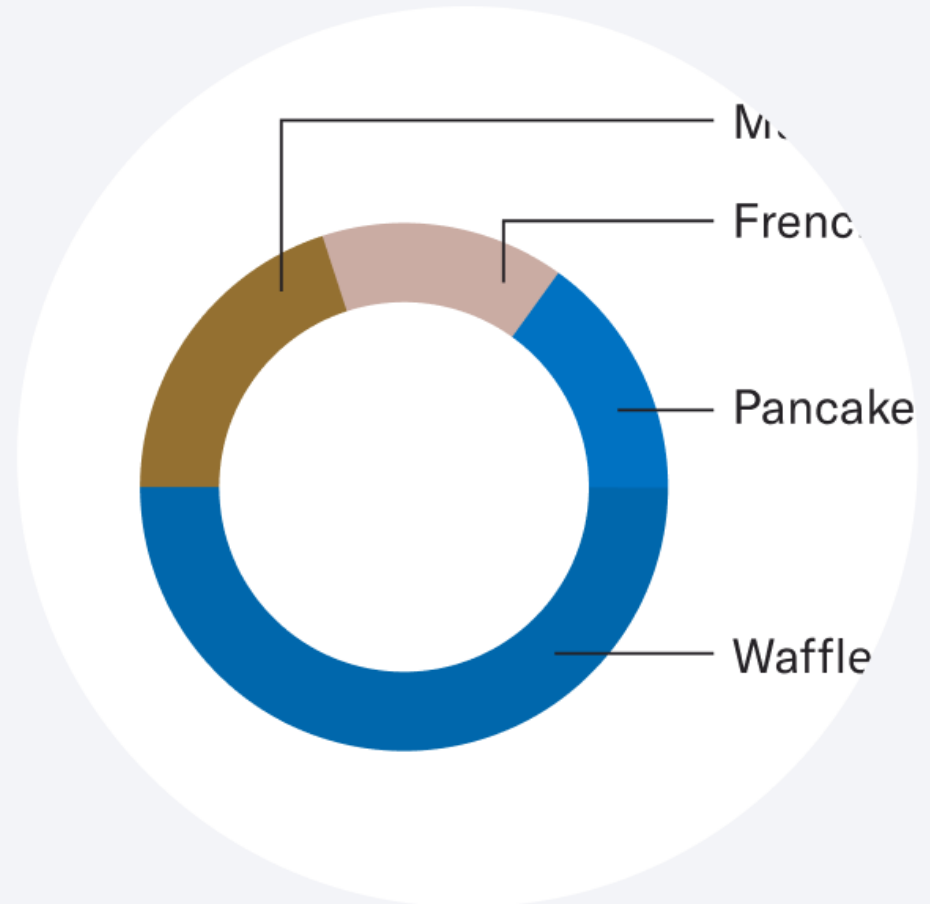


No filter applied

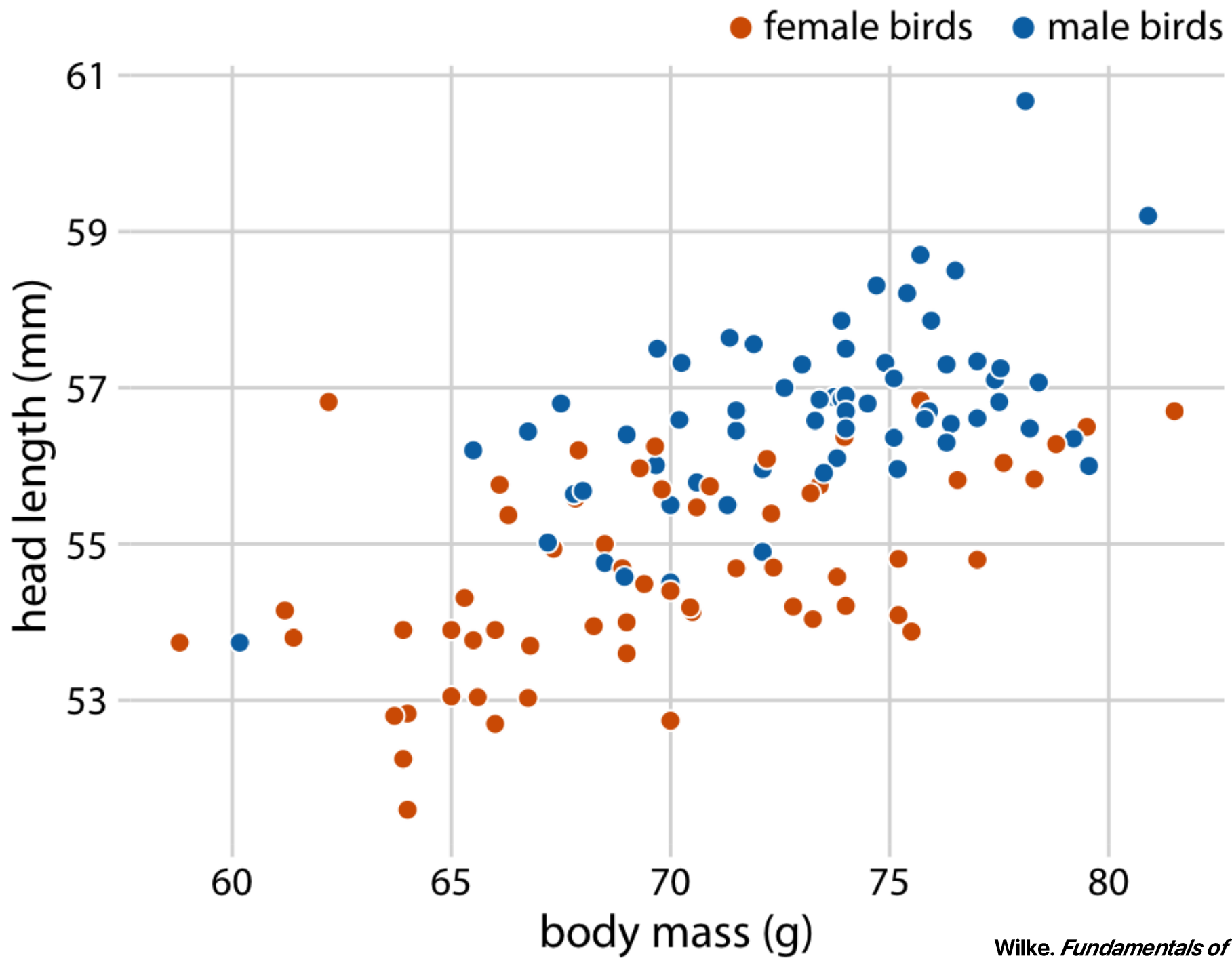
DO



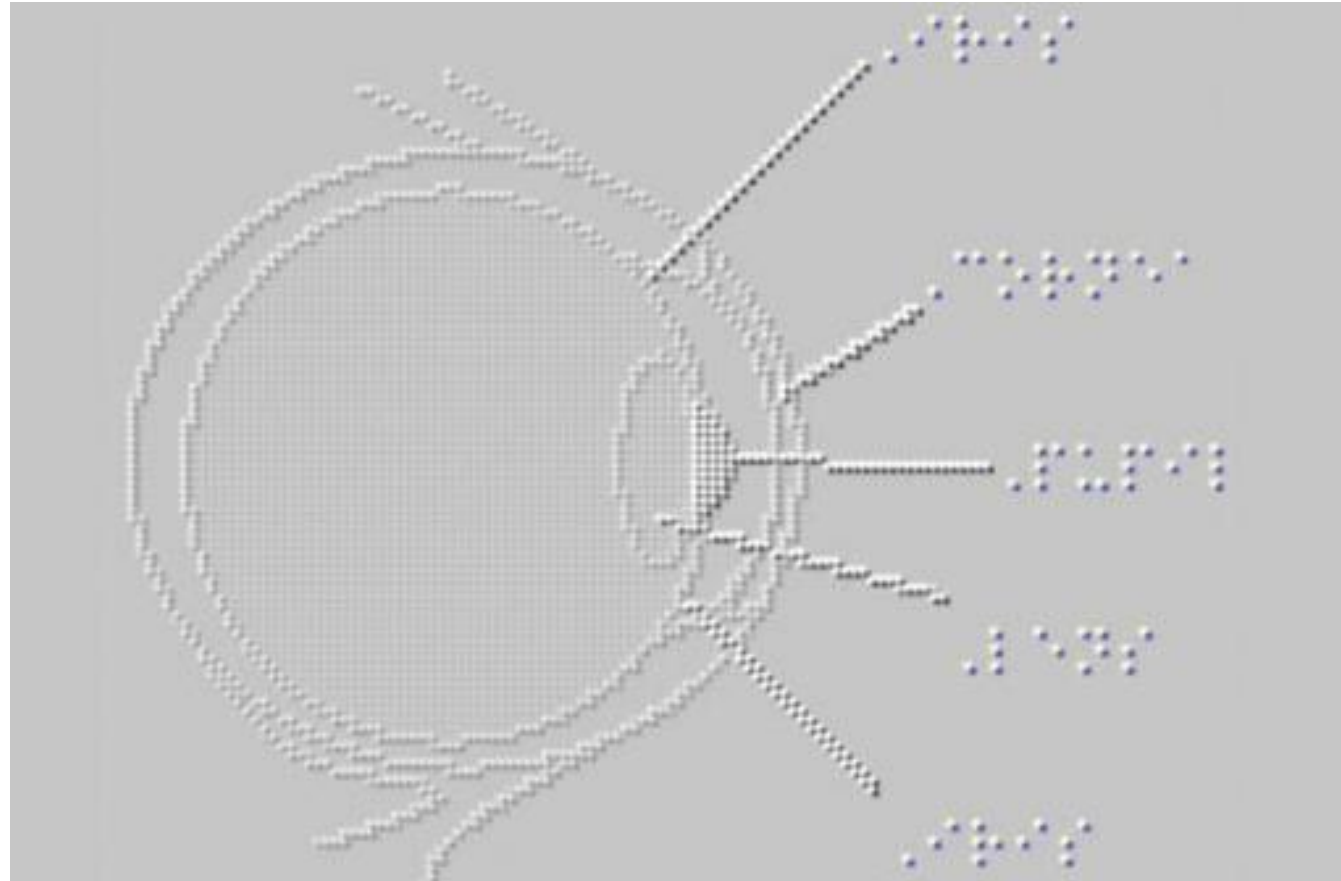
DON'T

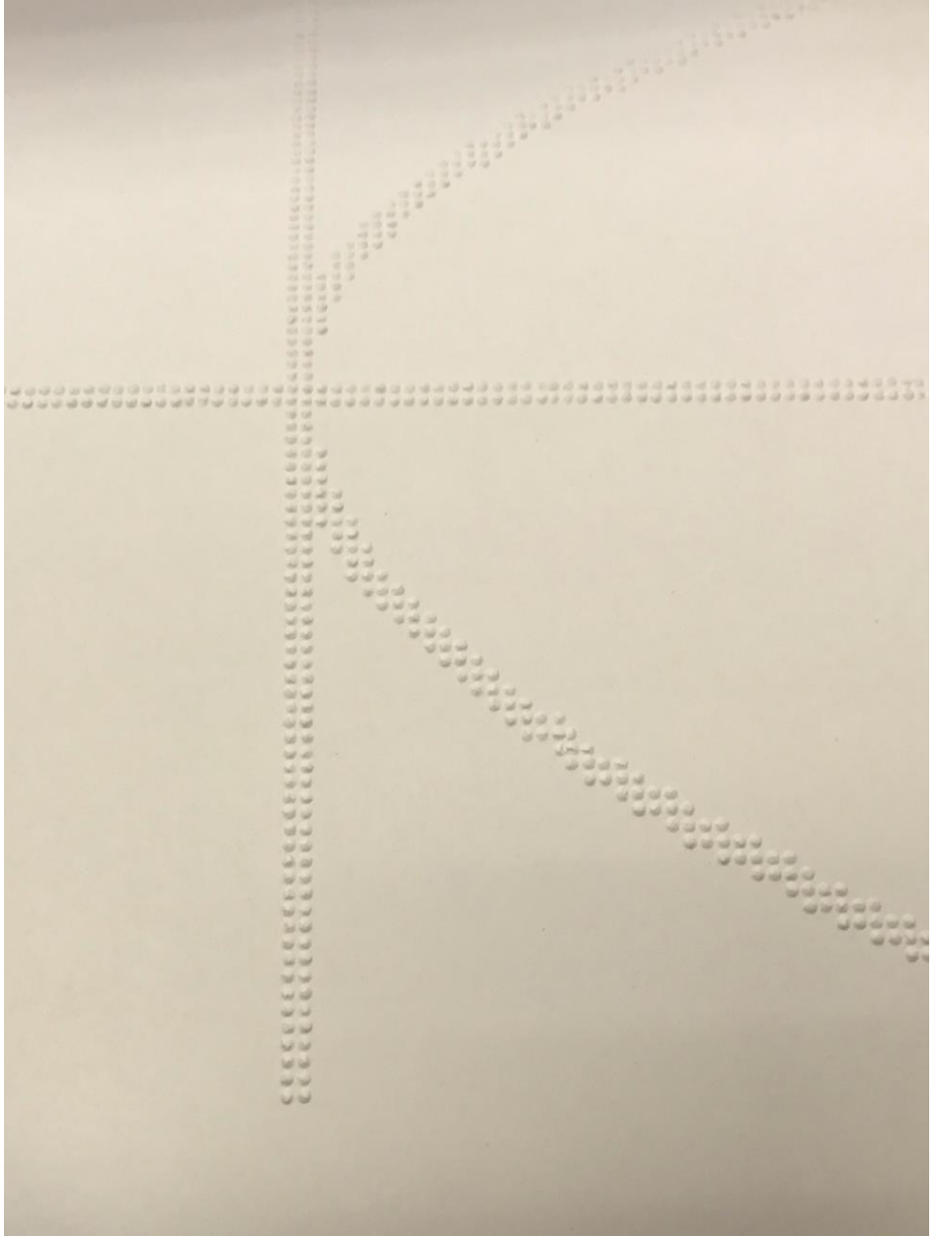
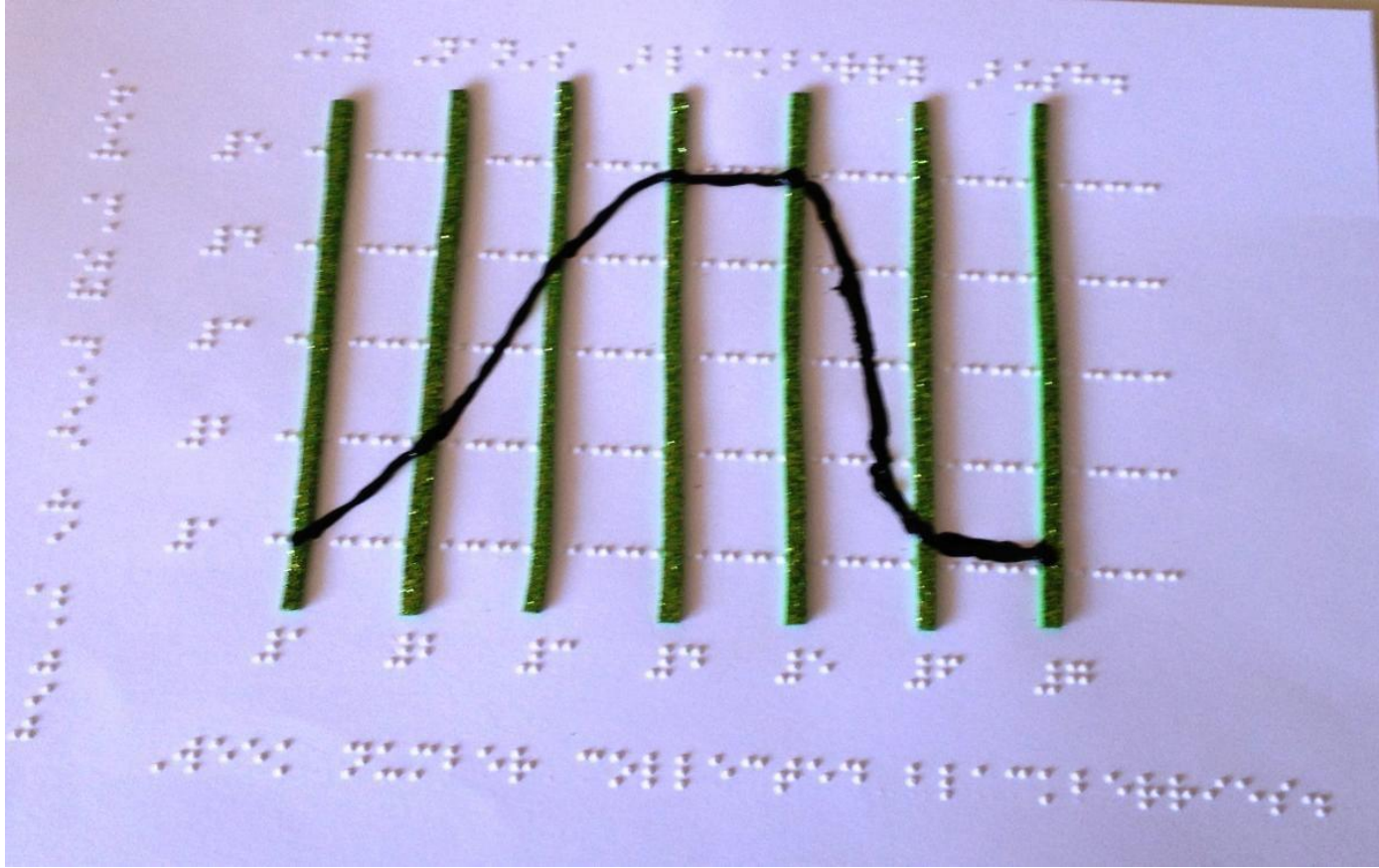


With Deuteranopia filter applied



Think about How Your Visualization Might Convert to a Tactile Graphic



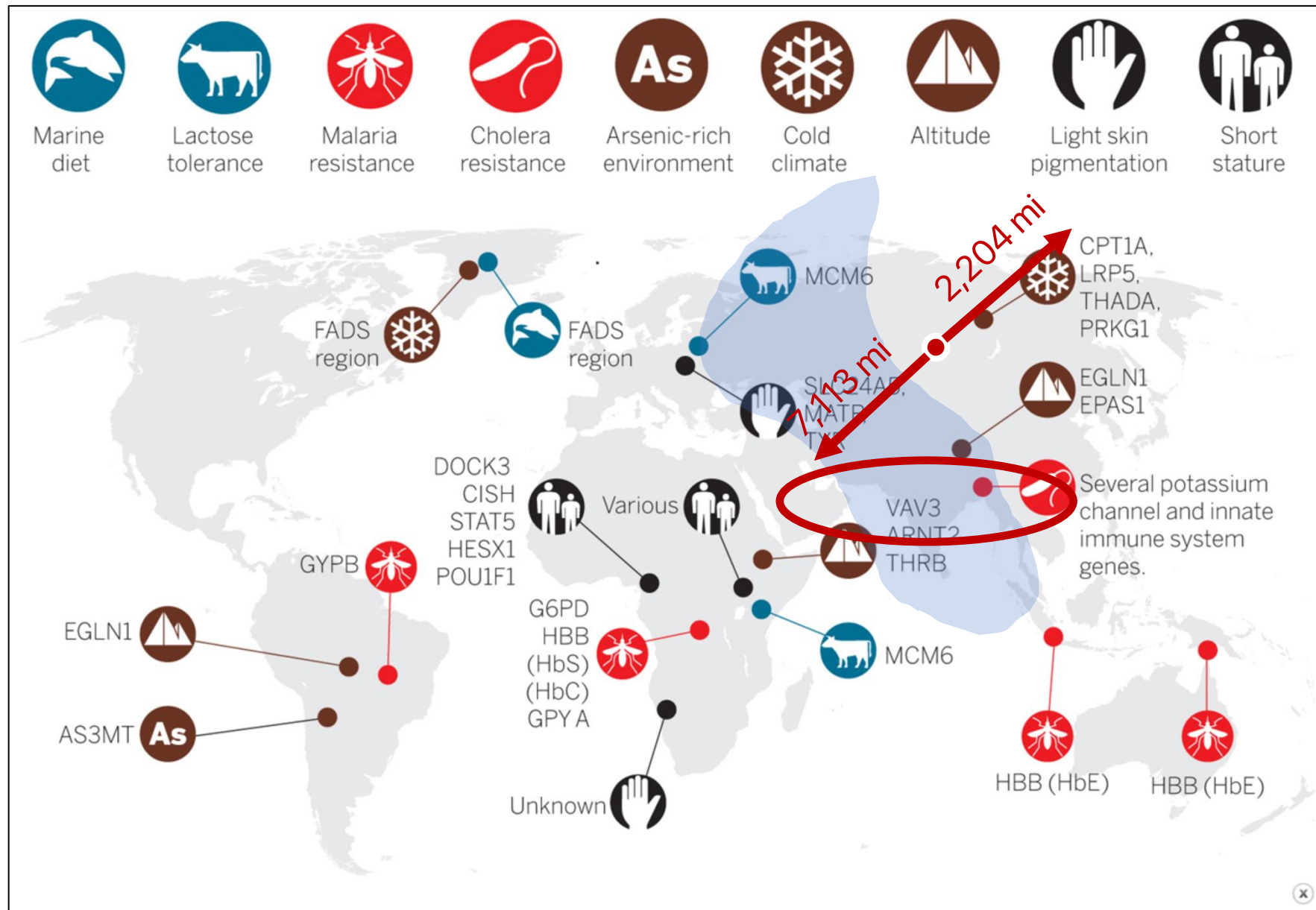


Accessibility

- For blind individuals in your audience, good **alt text** and a good **title** and **caption** will often serve best because the most common assistive technology will be a screen reader.
- A **refreshable brail display** enables blind readers to read without being read to.



Justice, Equity, Diversity and Inclusion



CREBRF





Justice, Equity, Diversity and Inclusion

- Use language that is **accessible and inclusive**
 - Person-first language: **people with disabilities** *vs* **disabled people**
 - Some communities prefer identity-first language, such as the blind and deaf communities
 - Generally, use racial and ethnic categories as adjectives and not as nouns: **Hispanic participants**, *not* **Hispanics**
 - Communicate with your audience ...
What is their **terminology for themselves**?
 - A helpful guideline: Flanagin *et al.* *JAMA* 2021 [↗](#)
 - The **language will change** as everyone's understanding of themselves and each other change

Justice, Equity, Diversity and Inclusion

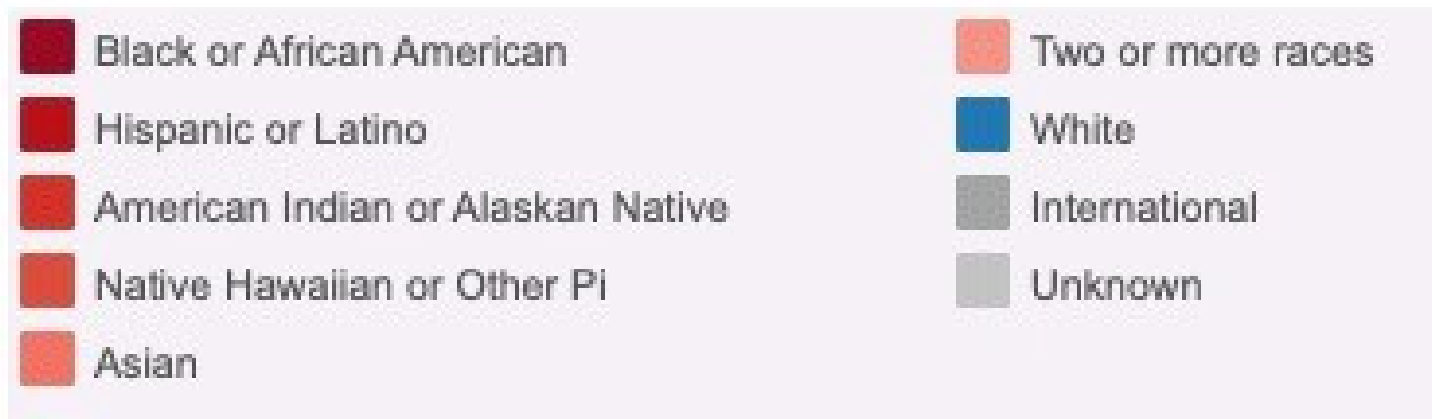
- Use language that is **accessible and inclusive**
- **Order labels intentionally**
 - Are you always putting "White" or "Men" first?
 - If your study is focused on a community, put that community first
 - Is the order consistent with your story or argument?
 - Is there a quantitative variable that serves to naturally order the groups? Alphabetically, population size, sample size, effect size?

Justice, Equity, Diversity and Inclusion

- Use language that is **accessible and inclusive**
- **Order labels intentionally**
- **Who is missing** from the visualization?
 - Is a racial or ethnic group absent?
 - Are non-binary individuals absent?
 - Do you describe who is in the group "Other"?

Justice, Equity, Diversity and Inclusion

- Use language that is **accessible and inclusive**
- **Order labels intentionally**
- **Who is missing** from the visualization?
- **Use color intentionally**
 - Avoid **Men** and **Women**, i.e. colors that reinforce stereotypes
 - How does your choice of colors foreground or background particular groups?



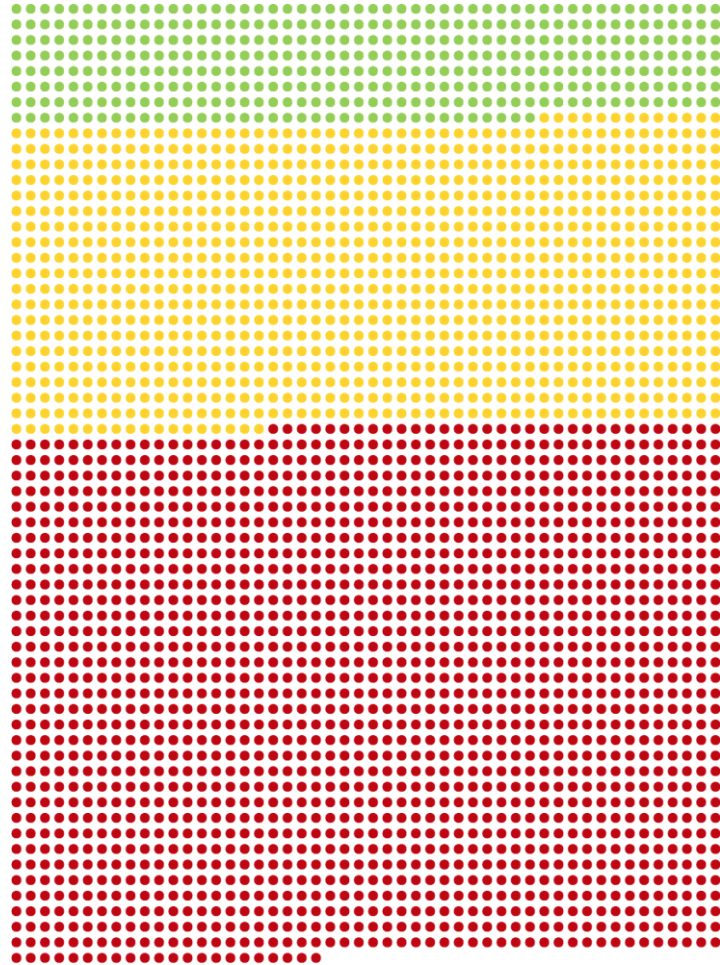
MIT Diversity Dashboard June 2020



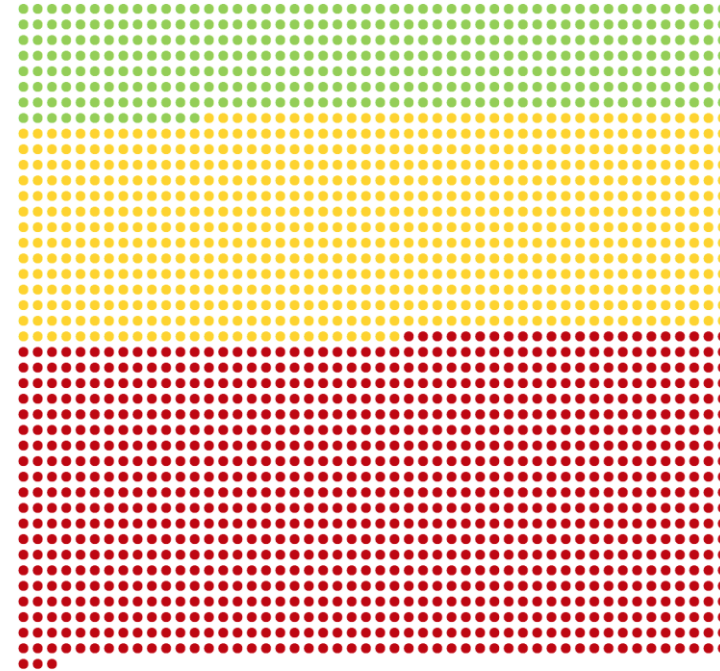
MIT Diversity Dashboard July 2020

Justice, Equity, Diversity and Inclusion

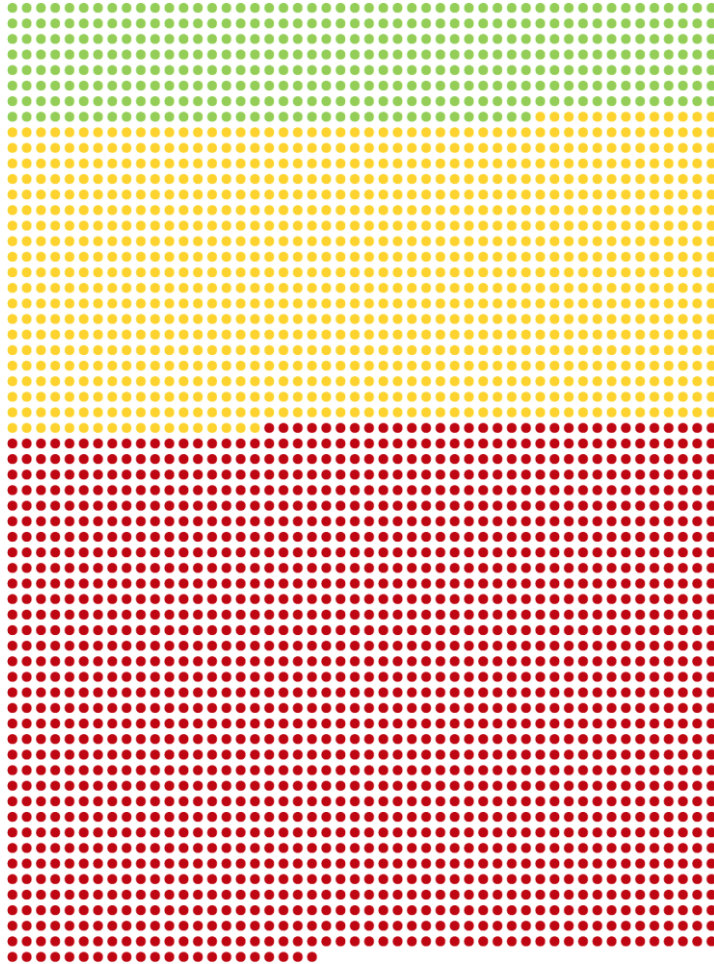
- Use language that is **accessible and inclusive**
- **Order labels intentionally**
- **Who is missing** from the visualization?
- **Use color intentionally**
- Consider **humanizing your visualizations**
 - Can you foreground the people in your visualizations?



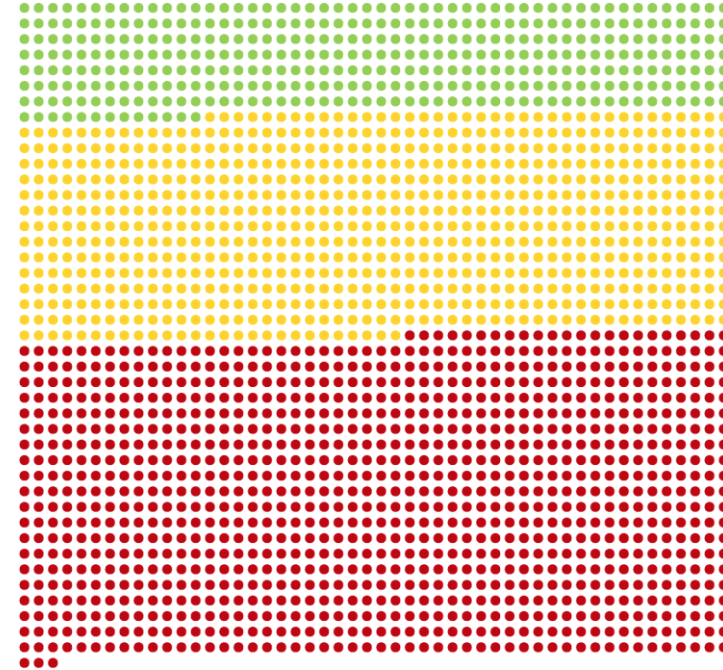
Discovery $n = 3,072$
with Normal Weight · 12.6 %
with Overweight · 31.9 %
with Obesity · 55.5 %



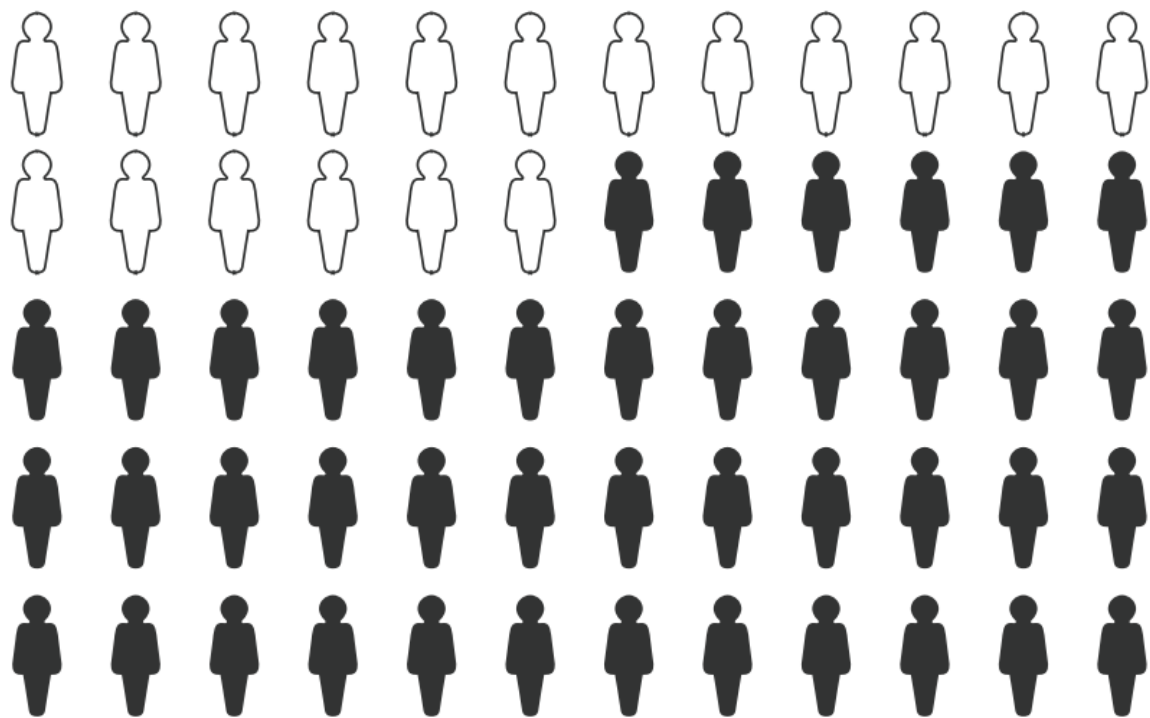
Replication $n = 2,103$
with Normal Weight · 17.3 %
with Overweight · 34.0 %
with Obesity · 48.8 %



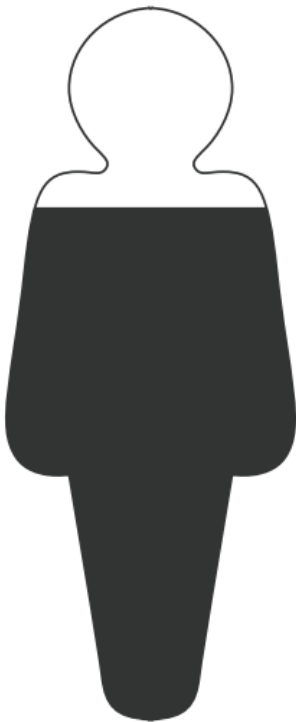
Discovery $n = 3,072$
 $< 26 \text{ kg/m}^2 \cdot 12.6 \%$
 $\geq 26 \text{ kg/m}^2 \text{ \& } < 32 \text{ kg/m}^2 \cdot 31.9 \%$
 $\geq 32 \text{ kg/m}^2 \cdot 55.5 \%$



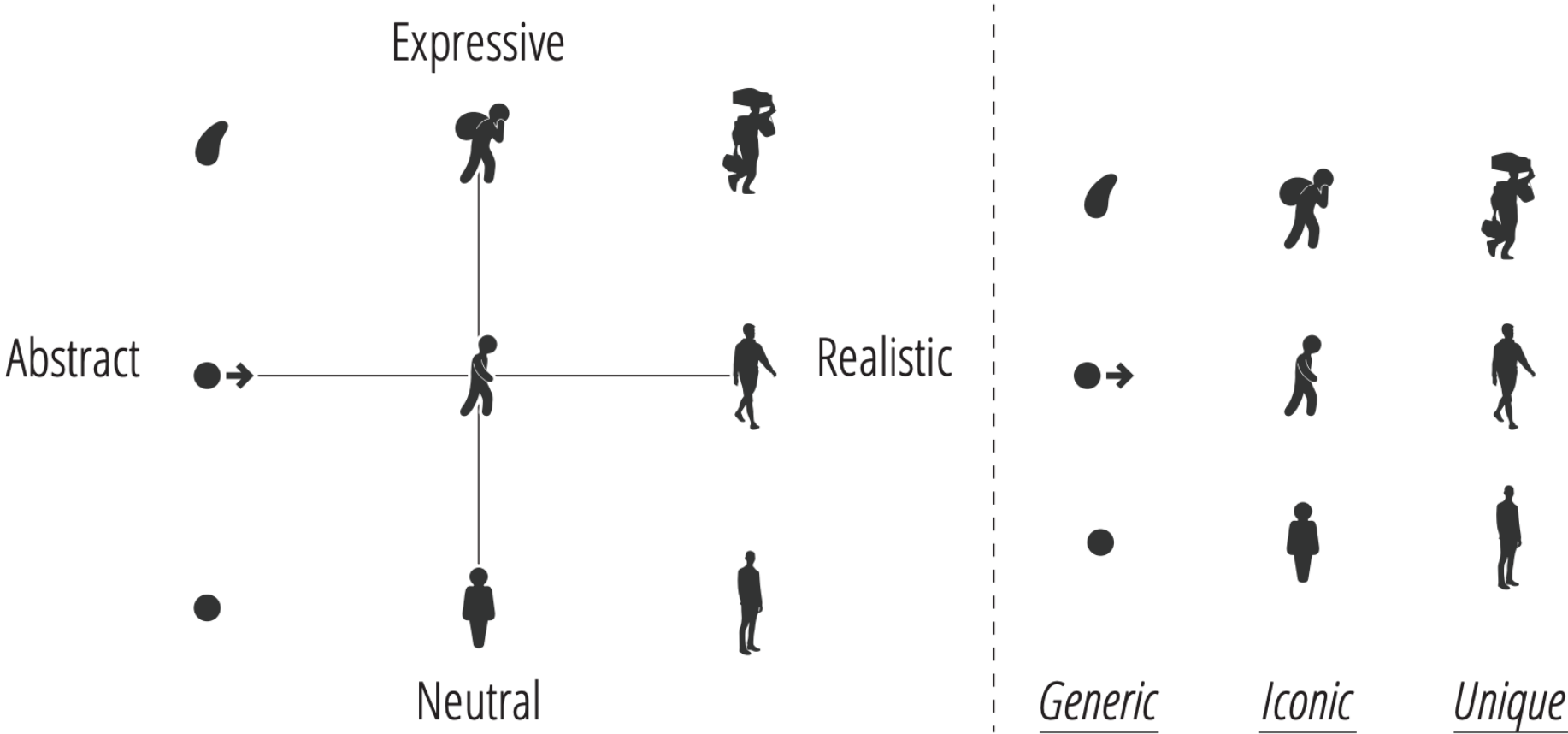
Replication $n = 2,103$
 $< 26 \text{ kg/m}^2 \cdot 17.3 \%$
 $\geq 26 \text{ kg/m}^2 \text{ \& } < 32 \text{ kg/m}^2 \cdot 34.0 \%$
 $\geq 32 \text{ kg/m}^2 \cdot 48.8 \%$



Unit



Aggregate





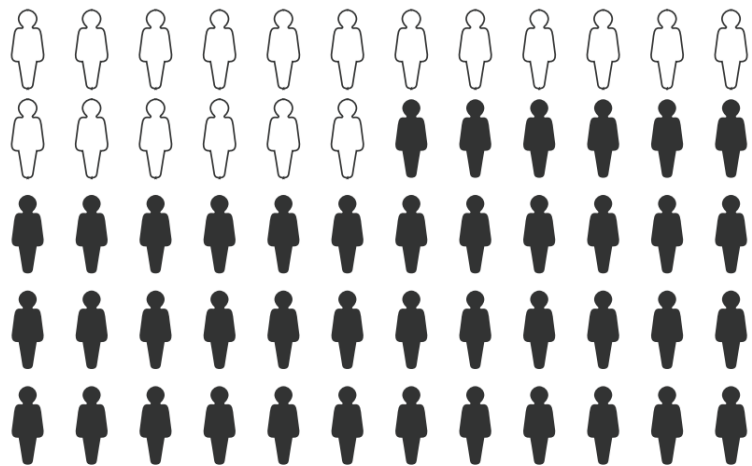
Generic



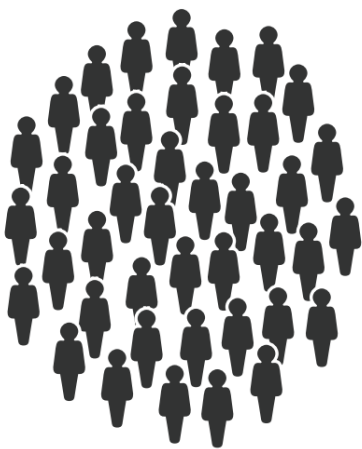
Iconic



Unique



Grid

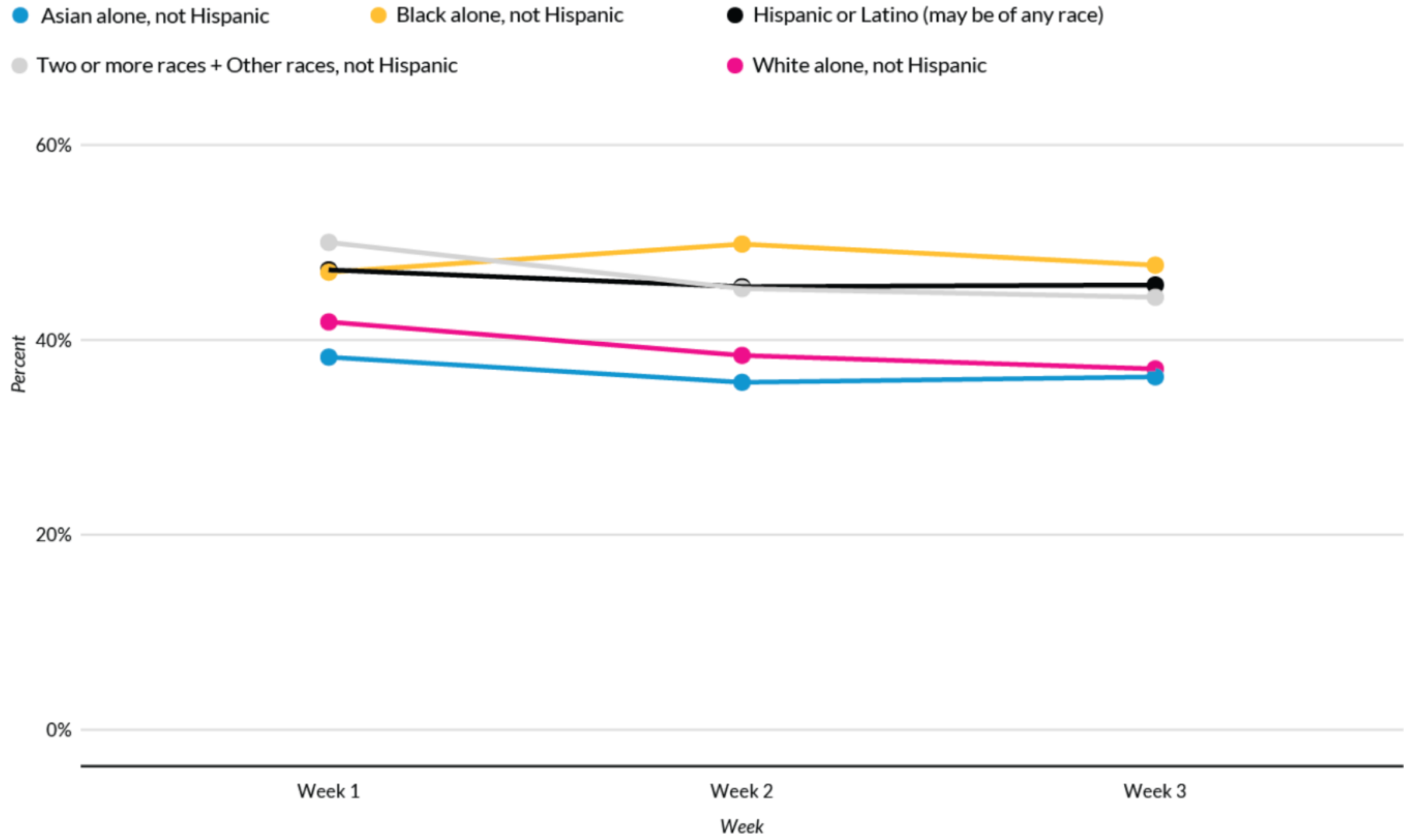


Organic



Justice, Equity, Diversity and Inclusion

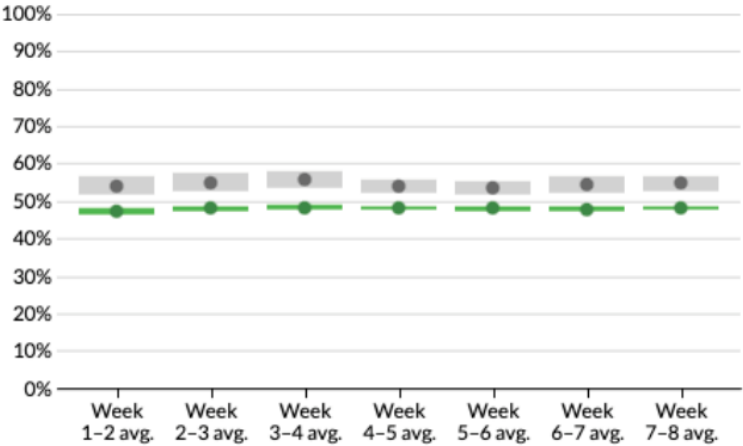
- Use language that is **accessible and inclusive**
- **Order labels intentionally**
- **Who is missing** from the visualization?
- **Use color intentionally**
- Consider **humanizing your visualizations**
- Are you inviting **unnecessary or inappropriate comparisons**?
 - Instead of comparing groups against each other, can you compare them to an average or a target?



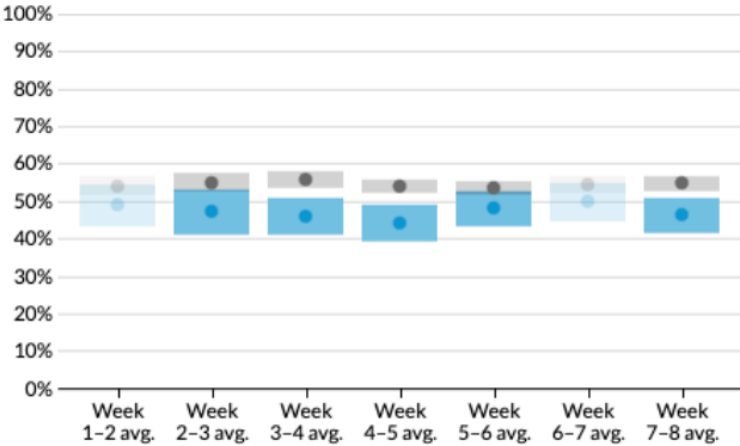
Share of adults in households where at least one person has lost employment income since March 13



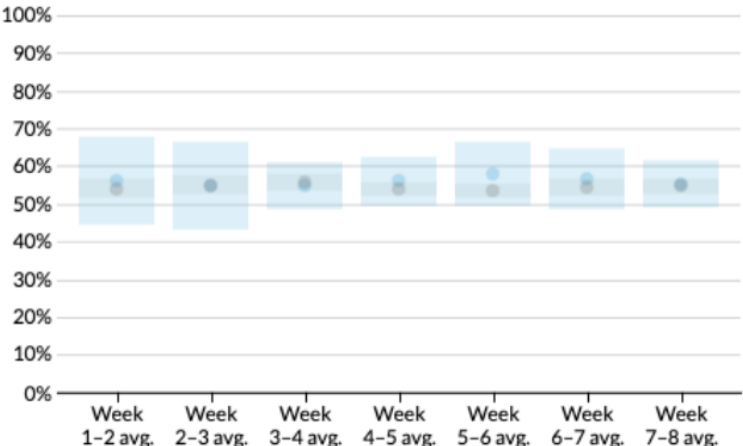
All



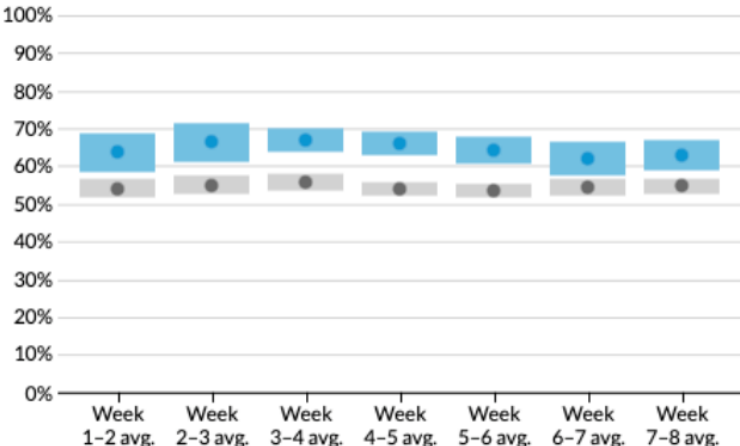
Asian



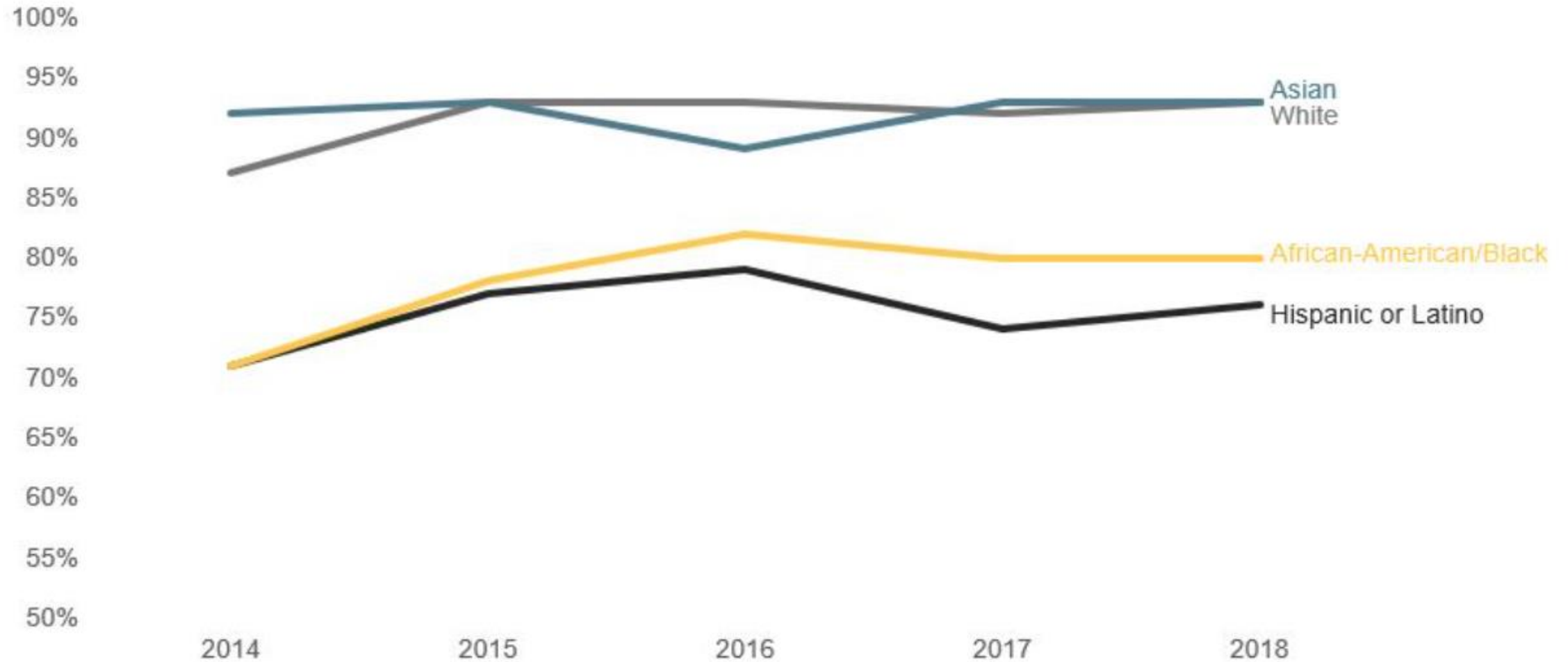
Black



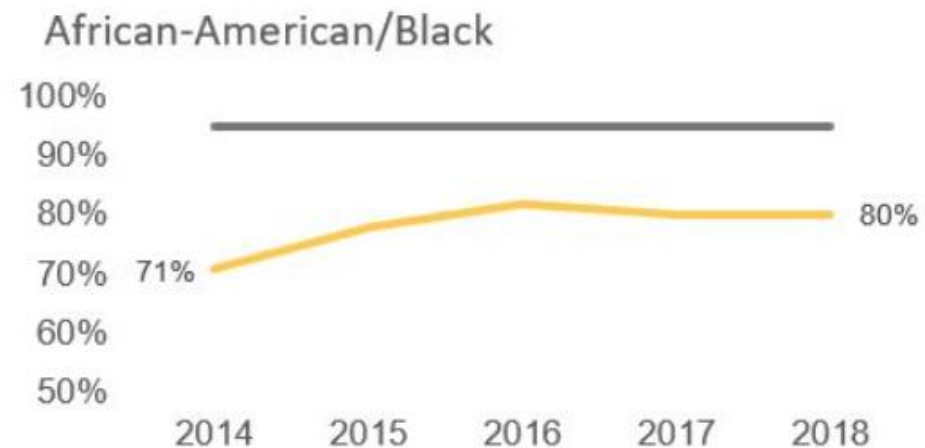
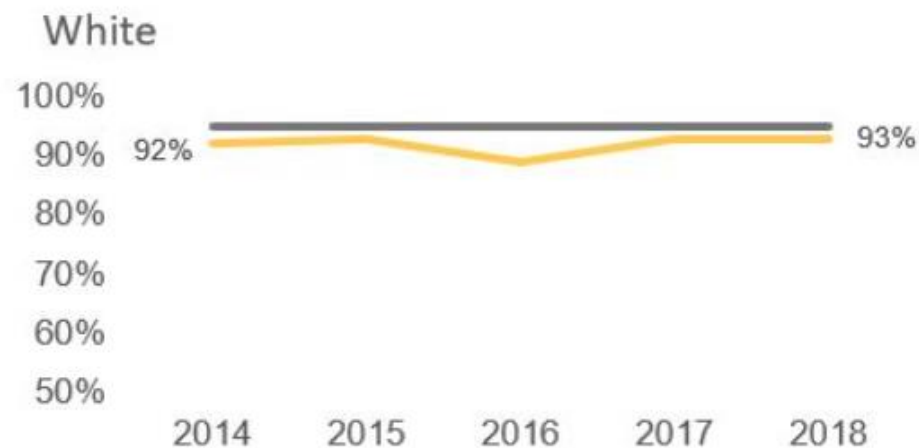
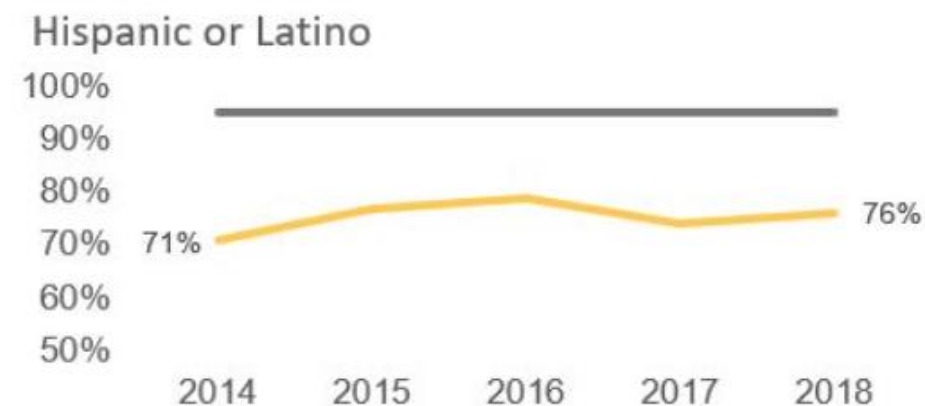
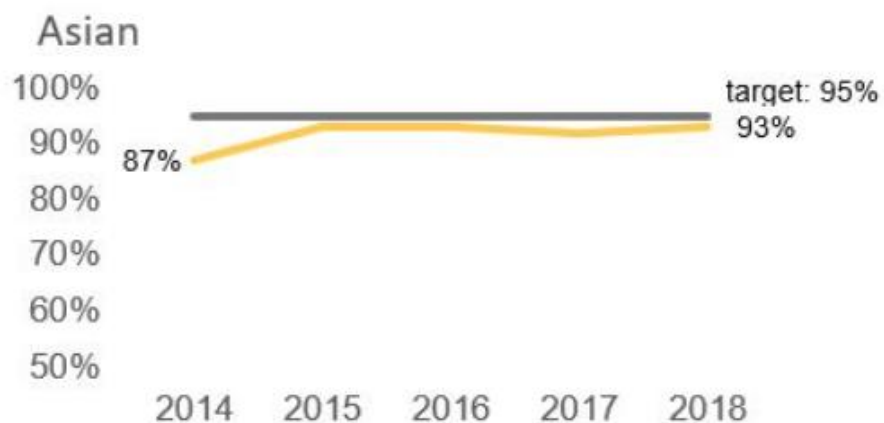
Hispanic or Latino



Boston 10th graders scoring proficient or higher on Massachusetts Comprehensive Assessment System



Boston 10th graders scoring proficient or higher on Massachusetts Comprehensive Assessment System



Ethics

- **Veracity** truth, accuracy
- **Fidelity** integrity in relationships, keeping promises
- **Autonomy** self-agency, ability to make decisions for oneself
- **Justice** fairness, equity, non-discrimination, transparency
- **Nonmaleficence** avoid harm
- **Beneficence** do good